



**UNIVERSITY
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**A Framework for Predicting Retrieval-Augmented Generation System
Performance Based on Relevant Factors**

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Abstract

Retrieval-augmented generation (RAG) systems address a fundamental limitation of large language models by grounding responses in externally retrieved content rather than relying solely on parametric knowledge. While RAG adoption has expanded considerably across academic, commercial, and public-sector applications, organizations implementing these systems face persistent uncertainty about how knowledge base properties affect operational performance. In particular, the quantitative relationship between dataset freshness, source type diversity, and system performance across domains with different information volatility characteristics remains poorly understood.

This study addresses four objectives: to analyze how dataset freshness and source type relate to RAG performance across high-, medium-, and low-volatility domains; to examine the individual contribution of source type configurations to multiple performance metrics; to develop a predictive framework that estimates RAG effectiveness from dataset-level factors; and to evaluate the framework against benchmark performance ranges across diverse configurations.

A quasi-experimental design is employed, constructing twelve conditions by crossing four freshness levels with three source type configurations across three domains: technology (high volatility), healthcare (medium volatility), and historical research (low volatility). Each condition is evaluated using 200 queries against primary metrics of retrieval precision, response accuracy, and hallucination rate. Statistical analysis combines two-way ANOVA, multiple regression, and Random Forest modelling with ten-fold cross-validation.

The primary contribution is a validated predictive framework that translates dataset characteristics into expected performance ranges with documented boundary conditions. This enables evidence-based decisions about knowledge base update frequency and source type selection without requiring prior experimental data specific to a given deployment context. The findings have direct practical implications for organizations seeking to optimize RAG knowledge base composition and maintenance strategies in resource-constrained settings.

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CHAPTER 1: INTRODUCTION

1.1 Research Background

Language models continue to advance, yet fundamental limitations persist. Traditional large language models rely exclusively on knowledge acquired during training. Once the training window closes, their knowledge becomes fixed. When queried about recent events, these models face inherent challenges in providing accurate information. They may generate plausible-sounding responses, but without access to current data, such responses are essentially produced from static, outdated knowledge. This limitation has motivated researchers to reconsider how systems access and use information.

Retrieval-augmented generation emerged as a response to this constraint. The core concept is straightforward: rather than relying solely on static weights from training, the model retrieves external context to ground its generation. When processing a query, the system first searches through external documents to identify relevant information, then uses that retrieved content to generate a response. While conceptually simple, optimizing this architecture has proven complex. Lewis et al. (2021) conducted foundational work demonstrating that this approach could improve factual accuracy by approximately 9% to 19%, depending on the task. Beyond accuracy gains, the key advantage lies in the ability to update external knowledge bases without retraining entire models.

RAG system performance depends on multiple factors that influence how effectively these systems retrieve and use external knowledge. These factors span different levels: model architecture choices, retrieval algorithms, generation strategies, and knowledge base characteristics. Among these, dataset-level factors—properties of the external knowledge base itself—represent a particularly important category. Unlike model-specific optimizations that require retraining or architectural changes, dataset-level factors are stable across different model implementations, making them model-agnostic. They are also directly controlled by system

designers through knowledge base curation and maintenance decisions. This controllability makes dataset-level factors particularly suitable for practical optimization, as organizations can adjust them without modifying underlying model architectures.

The operational mechanism involves two distinct steps. First, the system retrieves relevant passages from a document collection using dense retrieval methods—finding documents whose vector representations closely match the query in high-dimensional space. Subsequently, it feeds both the original question and retrieved passages to a generation model, which produces the final response. The generation model can ground its answer in concrete sources rather than attempting to recall everything from parameters. In theory, this technique should reduce hallucinations. In practice, effectiveness varies based on multiple interacting factors.

What determines RAG system effectiveness depends heavily on retrieval source quality. The quality of the knowledge base matters substantially. Wang et al. (2022) investigated how diverse types of external knowledge influence performance, comparing parametric knowledge—information encoded in model weights—with non-parametric approaches where knowledge resides in external databases. Their findings indicated that high-quality sources could improve accuracy by approximately 27%, but improvements did not continue linearly. Eventually, benefits from adding more high-quality sources began to diminish. The relationship between source quality and performance appears more complex than simple linear scaling.

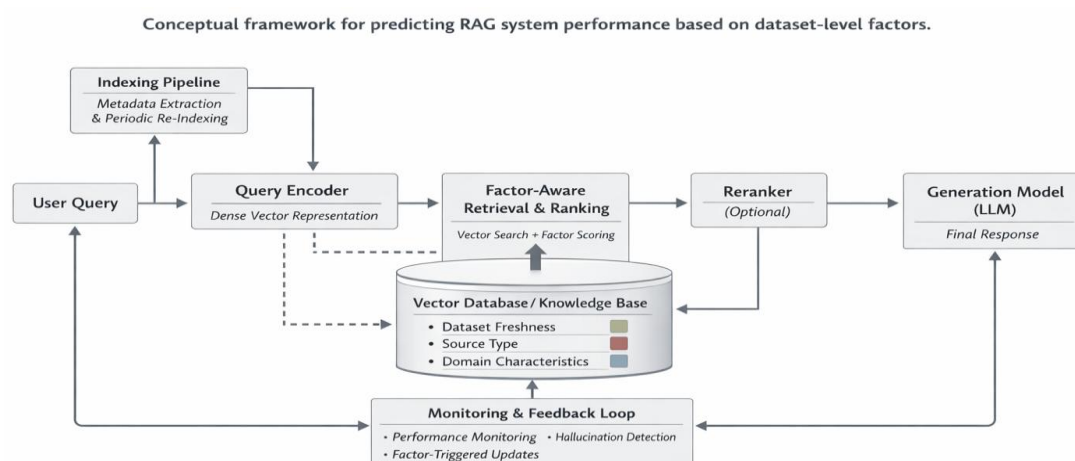


Figure 1.1: Conceptual framework for predicting Retrieval-Augmented Generation (RAG) system performance based on dataset-level factors.

Figure 1.1 illustrates the predictive framework architecture, emphasizing how dataset-level factors directly influence RAG system performance. The framework transforms user input into dense vector representations, which are compared against a database containing document embeddings and metadata representing key dataset factors (source type classification, temporal freshness indicators, domain characteristics). An indexing pipeline processes documents and extracts factor metadata before database insertion, with reindexing frequency determined by factor-specific requirements.

The framework employs vector similarity combined with factor-aware scoring to rank documents and identify top-k most relevant results. An optional reranker further refines results before passing them to the generation model, which produces the final response. A feedback and monitoring loop tracks retrieval relevance, detects hallucinations, and triggers updates when factor values threaten reliability. This architecture demonstrates the predictive framework's core principle: systematically quantifying how dataset-level factors—illustrated here through temporal freshness and source type as representative examples—affect retrieval quality, ranking accuracy, and overall RAG performance. The framework enables prospective prediction of system behaviour based on factor configurations before deployment.

Information freshness represents one critical dataset-level factor. Information ages at varying rates across domains. In rapidly evolving fields like technology or current events, information from last month may no longer apply today. Zhang et al. (2022) measured this issue by monitoring RAG performance decline as knowledge bases became outdated. Without updates, they observed a 2.3% monthly decline in technology-related tasks. While individual instances may seem minor, cumulative effects prove significant. After six months without updates, performance drops noticeably. The challenge is that different domains age at different rates. Historical facts remain valid indefinitely. Technology news becomes outdated within weeks or days.

Izacard et al. (2023) extended this analysis into few-shot learning scenarios where models work with limited examples. They conducted benchmarks across twenty different tasks and found that recent sources improved performance by approximately 17% on time-sensitive questions. Their work demonstrated that freshness affects more than factual recall—it also influences how effectively models reason about contemporary contexts and apply knowledge appropriately. However, they did not develop predictive frameworks that would allow practitioners to determine in advance how fresh a knowledge base needs to be for a given application.

Source type introduces additional complexity as another dataset-level factor. Documents vary in quality, style, and structure. Academic papers follow distinct conventions. News articles use different formats. Technical documentation occupies another category entirely. Each type brings different strengths. Gao et al. (2023) examined zero-shot retrieval across heterogeneous collections and found that retrieval precision varied substantially depending on source type. Academic sources appeared to boost factual precision. News articles enhanced contemporary relevance. Technical documents improved accuracy in specialized domains. However, optimal ratios for mixing these types and their intended purposes remain unclear.

Chen et al. (2022) observed interesting patterns regarding explainability. Users appear to trust RAG systems more when they receive information from diverse source types. In their study, source variety increased explainability ratings by 41%. Users found it easier to verify claims when responses cited multiple source types. This proves important for real-world deployment, particularly in domains where accountability matters. People want to know information provenance, and source diversity appears to enhance confidence, even when it does not necessarily improve accuracy.

Evaluation presents challenges because two separate processes—retrieval and generation—interact in complex ways. Mallen et al. (2023) made an important observation: approximately 28% of detected hallucinations originated from retrieved content itself, not from generation processes. This finding challenges the assumption that hallucinations primarily reflect generation problems.

Occasionally, the retrieval step identifies poor or outdated information. The generator then faithfully reproduces those errors. Regardless of generation model quality, it cannot address problems originating in retrieved content.

Recent work has attempted to incorporate self-correction mechanisms. Asai et al. (2023) introduced SELF-RAG, which adds reflection tokens enabling models to assess whether retrieved passages actually help answer questions. Instead of blindly using whatever retrieval returns, models can evaluate relevance and ignore unhelpful passages. This approach showed improvements, though it added computational overhead. Self-reflective components of this kind have been shown to reduce hallucinations by approximately 32% compared with standard RAG. The model essentially validates its own work before committing responses.

User satisfaction does not always align with expected patterns. Xiong et al. (2024) conducted a mixed-methods study combining technical metrics with actual user feedback. They found that users often value source diversity over strict accuracy. People appreciated comprehensive responses presenting multiple perspectives, even if individual sources contained minor errors. This suggests that optimizing purely for accuracy might overlook important aspects of how users perceive quality and usefulness.

Despite substantial research, the field presents a fragmented picture. Studies often focus on specific aspects—individual factors like freshness, source type, or evaluation methodology—without fully integrating these into predictive frameworks. Organizations implementing RAG systems face practical questions: How often should knowledge bases be updated? What mix of source types should be included? How do these factors interact? Currently, these decisions often depend on intuition or available resources rather than empirical evidence. The field requires systematic investigation of how dataset-level factors combine to influence performance and, critically, frameworks that enable predicting outcomes before deployment.

1.2 Problem Statement

RAG systems are being adopted rapidly across companies, universities, and public services. However, a significant gap exists between the scale of their deployment and the understanding of how controllable, dataset-level factors influence operational performance. This disconnect creates real challenges for anyone trying to implement these systems thoughtfully.

1.2.1 Resource Allocation Challenges

Resource allocation presents immediate challenges. Maintaining knowledge bases requires computational resources for indexing and searching documents, as well as personnel to curate sources, verify quality, and manage updates. Organizations must decide: Should knowledge bases be updated daily? Weekly? Monthly? Which source types should be prioritized? What return on investment can be expected?

Lewis et al. (2021) acknowledged that maintenance costs are significant but did not establish clear relationships between factor configurations—such as update strategies and source selection—and performance improvements. Without such relationships, organizations essentially guess whether resource expenditure is worthwhile. A small startup might lack the budget for daily knowledge base refreshes, but how much performance is sacrificed by updating weekly instead? These questions arise constantly in practice, yet solid answers remain elusive.

1.2.2 Domain-Specific Variations

Different domains exhibit different requirements regarding dataset-level factors, though boundaries remain unclear. Technology evolves rapidly. Historical research changes slowly. But what about healthcare? Legal information? Financial news? Unclear boundaries complicate planning.

Asai et al. (2023) demonstrated that selective retrieval—targeting relevant content rather than using everything available—could optimize performance-resource tradeoffs. However, their work focused on model-level self-reflection, and generalising findings on optimal knowledge base update strategies to different domains remains unclear. A medical research organization likely

has different factor requirements than a technology news aggregator, but systematic principles enabling organizations to assess domain-specific factor configurations remain lacking. Without such principles, organizations make educated guesses based on intuition rather than evidence.

1.2.3 Source Type Selection Complexity

Choosing source type configurations presents similar challenges. Academic papers offer high quality but lag current developments. News articles provide timeliness but vary in reliability. Technical documentation offers precision for specific domains but narrow coverage. Web content provides abundance and diversity but includes substantial noise. Each option involves tradeoffs, making optimal combinations difficult to determine.

Wang et al. (2022) found that source quality matters but observed non-linear relationships where adding more sources eventually stops helping. The mechanisms underlying this remain poorly understood. At what point does adding sources merely introduce redundancy without improving coverage? How should different source types be balanced for different use cases? For customer support RAG systems, should technical documentation be weighted more heavily than general web content? These practical questions lack clear answers.

1.2.4 Interaction Effects Between Factors

Dataset-level factors do not operate independently, creating additional complexity. Temporal freshness and source type interact in non-obvious ways. Recent academic papers might be ideal for emerging research topics, but academics do not publish on breaking news. News sources cover current events but lack peer-reviewed research depth. Technical documentation remains relevant longer in stable systems but becomes outdated quickly in rapidly evolving technologies. These factors interact, but interactions have not been mapped systematically. Fresh news articles might compensate for older academic papers in some domains, or they might serve entirely different purposes. Perhaps combining diverse source types matters more in some contexts and

less in others. Most research examines these factors in isolation, which does not reflect how they actually work together in real implementations.

1.2.5 Evaluation Framework Limitations

Current evaluation methods treat RAG systems as opaque entities. Measurements capture overall performance without isolating which factors contribute to problems. Mallen et al. (2023) noted something striking: approximately 78% of implementation failures they examined stemmed from dataset problems—factor misconfigurations—not algorithmic limitations. This is a remarkable finding.

It suggests research attention has focused on the wrong area—optimizing algorithms and architectures while real issues lie in knowledge base construction and maintenance, which are factor-level decisions. Evaluation frameworks capable of separating factor effects from model effects remain underdeveloped. When RAG systems underperform, is it because retrieval algorithms need tuning? Because generation models are struggling? Because dataset-level factors are misconfigured? Current methods make this difficult to determine.

1.2.6 Performance Decay and Predictability

Performance decay over time presents another area with more questions than answers. Zhang et al. (2022) quantified decay rates in specific domains—that 2.3% monthly decline in technology tasks—but did not build models enabling prediction of when factor-driven performance will hit unacceptable levels. Organizations are left conducting periodic manual checks or hoping their systems remain accurate enough. Both approaches scale poorly and undermine user confidence in information reliability.

From an operational perspective, organizations running production RAG systems need to know when performance will degrade enough to require factor adjustments, updating datasets, rebalancing source types. How much buffer should be built in? Can we predict which knowledge base components will age fast based on factor analysis? Currently, these remain open questions.

Organizations either overinvest in updates as precautionary measures or neglect them, identifying issues only when users complain.

1.2.7 The Need for Integrated Frameworks

The fundamental issue is the absence of an integrated predictive framework grounded in relevant dataset-level factors. By “relevant factors,” this study refers to dataset-level properties that are: (1) stable across model architectures, (2) directly controllable through knowledge base management decisions, and (3) demonstrably influential on system performance. Examples include the temporal freshness of documents, source type composition, domain coverage, and information granularity. This research focuses on dataset freshness and source type as representative relevant factors—chosen because they reflect all three characteristics (stable, controllable, influential) and appear consistently in the literature as performance determinants yet have never been systematically integrated into a predictive framework.

This study builds a predictive framework that quantifies factor-performance relationships, not an exhaustive catalogue of all possible factors. The framework approach is generalizable: while demonstrated through two factors, the methodology could extend to other dataset-level properties. This design choice reflects practical constraints—comprehensive factor analysis would be intractable—and research priorities—establishing proof-of-concept for factor-based prediction before expanding scope.

Organizations need to forecast how different factor configurations will affect performance. They need to understand not just averages but variance across domains and use cases. Organizations require tools enabling evidence-based decisions about knowledge base composition—essentially, factor optimization—rather than relying solely on intuition.

Research Gap Summary: Current literature examines dataset-level factors in isolation, provides limited temporal modeling of factor effects, lacks predictive frameworks for practical factor-based decision-making, and inadequately addresses interaction effects between factors. No existing

work systematically quantifies how factor combinations affect RAG performance across domains with different characteristics, nor provides validated predictive frameworks enabling organizations to optimize factor configurations based on empirical evidence rather than heuristics. The fragmentation across studies examining individual factors prevents integration into actionable predictive models.

This research aims to build such a framework. It will not resolve every open question in the field, but it can establish systematic, evidence-based foundations for factor-driven performance prediction. The goal is progress rather than perfection. If factor-performance relationships can be quantified, benchmarks established for different factor configurations, and predictive models built that hold up under validation, that represents a meaningful step forward from the current state of ad-hoc, intuition-driven decision-making.

1.3 Research Questions

The following research questions map directly to the four research objectives. Each question is framed with sufficient specificity to guide the experimental design and analysis described in Chapter 3. Together, they establish the scope and direction of the study.

1. How do dataset freshness and source type, as key dataset-level factors, relate to RAG system performance across high-, medium-, and low-volatility domains, measured as changes in retrieval precision, response accuracy, and hallucination rate?
2. What is the individual contribution of source type configurations to RAG system performance across multiple evaluation metrics, and does this contribution vary by domain volatility?
3. Can a predictive framework be developed that estimates RAG system performance from dataset-level factors, specifically freshness and source type, across domains with different volatility characteristics?

4. How accurately does the proposed predictive framework estimate RAG performance across different factor configurations and domain contexts, and what are its reliable operating boundaries?

1.4 Research Objectives

The following objectives directly support framework construction by establishing empirical foundations for factor-based performance prediction.

1. Analyze the key dataset-level factors and their relationship with RAG system performance across high-, medium-, and low-volatility domains.
2. Examine the impact of the relevant factors on the performance of RAG across multiple metrics.
3. Develop a predictive framework capable of estimating RAG effectiveness based on relevant factors such as dataset freshness and source type.
4. Evaluate the proposed predictive framework using benchmark performance ranges established across different factor configurations and domain contexts.

1.5 Scope of Study

Clear boundaries must be established to avoid scope creep and maintain focus on realistic accomplishments. This study focuses specifically on dataset-level factors and their predictive utility.

Domains: Three specific domains are examined, selected for their different information volatility characteristics. Technology represents high volatility—rapid change where yesterday's best practice becomes today's legacy approach. Healthcare represents medium volatility—new research emerges constantly, but clinical practices evolve more slowly. Historical research represents low volatility—facts about past events do not change, though interpretations may shift gradually. These three provide sufficient variation to test framework generalizability while keeping the study manageable.

Factor Focus: This research examines dataset freshness and source type as representative relevant factors. Many other dataset-level properties could influence performance—document length, writing style, information density, structural organization, citation networks, authoritativeness measures. However, not everything can be examined simultaneously. The two selected factors appear foundational and practically relevant enough to warrant dedicated attention while establishing proof-of-concept for factor-based predictive frameworks. Future work can extend to other factors once these are better understood.

Factors Explicitly Excluded: To isolate dataset-level factor effects, this study holds constant several other potential influences on RAG performance:

- **Model architecture variations:** The same base transformer architecture (e.g., T5 or comparable) will be used throughout. Different model sizes or architectures could affect performance, but examining these would conflate model effects with dataset factor effects.
- **Embedding model choices:** A single dense retrieval model (e.g., Dense Passage Retrieval or sentence-transformers) will be used consistently. Different embedding models produce different vector representations, but this study focuses on how dataset factors affect performance given fixed embeddings.
- **Prompt engineering:** Query formulation and prompt templates will be standardized. While prompt engineering significantly affects generation quality, varying prompts would obscure dataset factor contributions.
- **Reranking algorithms:** If reranking is used, the same algorithm will be applied across all conditions. Reranking strategies affect which retrieved documents are prioritized, but this study examines how dataset factors affect what is available to retrieve and rerank, not which reranking strategy works best.

This exclusion strategy reflects the research focus on dataset-level factors that organizations can control through knowledge base management decisions. By holding model-level choices constant, the study isolates factor effects and produces findings that generalize across different

model implementations. Organizations can apply framework insights regardless of their specific model choices.

Performance metrics: Primary emphasis focuses on retrieval precision, response accuracy, and hallucination rate because these directly affect user experience and reflect how well dataset factors support system operation. For a fuller picture, secondary metrics like relevance and completeness will be tracked, but analysis centers on those three main dimensions. Processing time matters for practical deployment and will be noted, though it is not the primary focus.

System architecture: Open-source transformer models will be used—Dense Passage Retrieval or similar for retrieval and T5 or comparable architectures for generation. This choice ensures reproducibility and accessibility. Validation with proprietary models may be necessary, but commencing with open methodologies appears more prudent.

What is excluded beyond factors: Several aspects are explicitly not addressed. Multimodal RAG systems combining text with images or video fall outside this scope. Conversational scenarios requiring context tracking across multiple turns would add complexity beyond current capacity. User interface design and interaction patterns, while important for deployment, are not part of this investigation. These are not value judgments—they are practical limitations. Boundaries must be established.

Time frame: Dataset construction draws on sources from 2015 to 2026. This provides nearly a decade of material to work with, sufficient to examine freshness effects meaningfully while remaining relevant to current systems. Queries focus on contemporary information needs rather than historical or hypothetical scenarios.

CHAPTER 2: LITERATURE REVIEW

2.1 Resource Allocation in Retrieval-Augmented Generation (RAG) Systems

Retrieval-augmented generation (RAG) system implementation requires careful allocation of computational, human, temporal, and financial resources. Organizations face critical decisions about resource distribution while lacking systematic frameworks for optimization. Understanding resource requirements establishes context for evaluating approaches to dataset-level factor management—specifically freshness and source type configurations that this research examines. Computational resources represent the most visible costs. Indexing transforms documents into searchable vectors, requiring substantial processing power. Lewis et al. (2021) reported that initial Wikipedia indexing consumed approximately 42 GPU hours. Retrieval operations add overhead—each query triggers similarity computations across potentially millions of vectors. Ram et al. (2023) found retrieval accounts for 40–60% of total inference time. Storage requirements scale with knowledge base size; dense vector embeddings (768-1024 dimensions per chunk) translate to gigabytes for modest collections. Organizations must also store original documents and metadata.

Human resources remain essential despite automation. Domain experts must identify appropriate sources, verify quality, and establish inclusion criteria. Wang et al. (2022) estimated expert curation contributed 30-40% of implementation effort. Quality control demands ongoing attention—periodic output sampling to detect drift and validate retrieval relevance. Maintenance activities (performance monitoring, degradation investigations, and corrective measures) consume significant personnel time. Xiong et al. (2024) found organizations spent an average of 10–15 hours per week on production RAG maintenance.

Temporal resources involve updating frequency decisions. More frequent updates maintain freshness but consume more resources. Zhang et al. (2022) documented that daily updating organizations spent 3-4x more on maintenance than monthly updates, achieving 15-20% better

accuracy on time-sensitive queries. The challenge lies in determining optimal frequencies for different content types and domains. Quantifying this relationship across volatility levels is a central concern of this study, addressed through controlled experimentation that isolates the effect of dataset age on system performance.

Financial implications stem from all resource types. Cloud computing expenses for moderate-scale deployments reach thousands monthly. Gao et al. (2023) reported organizations with 10M+ documents typically spent \$2,000-5,000 monthly on infrastructure alone. Personnel costs often exceed computational expenses. The cost-benefit calculus remains poorly understood—should startups allocate budgets to daily updates or better source curation? These questions lack evidence-based answers. Building a predictive framework that connects dataset configurations to expected performance outcomes directly addresses this problem, giving organizations a principled basis for resource decisions.

Several recurring problems affect resource allocation decisions in current practice. First, organizations lack predictive frameworks that would allow them to estimate resource requirements before deployment. Most learn through costly trial-and-error. Second, resource decisions are typically made independently—update frequency is chosen separately from source selection—despite the likelihood of interactions between the two. Third, no standardized methodologies exist for comparing resource efficiency across different approaches. These issues are particularly acute for organizations with limited budgets, who need principled guidance on where to invest scarce resources for maximum return. Current literature provides insufficient support for these practical decisions, and this gap is directly addressed by this study through the establishment of performance benchmarks across different configurations of dataset freshness and source type.

2.2 Critical Issues in RAG Implementation

Beyond resource allocation, RAG systems face specific challenges related to dataset-level factors within this study's scope: freshness and source type. This section analyses these issues systematically.

2.2.1 Dataset Freshness Issues

Information decay patterns and domain variation: Knowledge relevance degrades over time, but patterns vary substantially. Zhang et al. (2022) measured a 2.3% monthly decline in technology domains but negligible decay in historical research over six months. Kasai et al. (2023) identified three distinct patterns: gradual (steady decline), threshold (stable then sudden drop), and cyclical (periodic fluctuations). Technology exhibited gradual decay. Financial information showed threshold patterns—remaining accurate for weeks, then becoming unreliable after events. Medical research displayed cyclical patterns tied to conference publications. This variation underscores why the impact of dataset age needs to be examined across domains with genuinely different rates of information change, rather than assumed to be uniform.

Within-domain variation complicates matters further. Within healthcare, treatment protocols for rapidly evolving conditions require frequent updates, while anatomical references never need revision. Existing research provides domain-level generalizations but insufficient granularity for practical decisions. Organizations building healthcare RAG systems need to know which content types require weekly versus annual reviews. Current literature lacks frameworks for making these fine-grained determinations. Systematic quantification of freshness effects at the domain level is precisely what is needed to close this gap.

Update scheduling and predictability challenges: Determining when to update requires predicting when performance will degrade below acceptable thresholds. Zhang et al. (2022) quantified historical decay rates but did not develop prospective models. Organizations currently rely on either fixed schedules (arbitrary) or reactive approaches (updating after detecting

problems). Neither optimally balances freshness against resource costs, directly relating to resource allocation challenges (Section 2.1).

Predictability limitations compound these challenges. Publication dates serve as rough proxies for freshness, but they overlook important nuances—later work updates, corrects, or contextualizes documents, thereby altering their effective age. Mallen et al. (2023) proposed tracking entity-mention timestamps rather than document creation dates, improving temporal accuracy by 23%. However, this method requires substantial preprocessing and may not generalize across content types. Forecasting performance based on freshness configurations before deployment, rather than diagnosing problems after the fact, is one of the core capabilities this study works toward.

2.2.2 Source Type Selection Issues

Quality-timeliness tradeoffs: Different source types occupy different positions on the quality-timeliness spectrum. Academic papers undergo rigorous peer review, ensuring high quality, but lag current events by months. News articles provide immediate coverage but vary wildly in accuracy. Technical documentation offers precision for specific domains but narrow scope. Organizations must balance these tradeoffs without clear guidance about optimal configurations. Understanding how each source type contributes to system performance across multiple evaluation dimensions is therefore an important part of this study.

Wang et al. (2022) found high-quality sources improved accuracy by 27%, but improvements plateaued beyond certain thresholds. This non-linear relationship suggests diminishing returns—adding more academic papers beyond some point provides little benefit. However, identifying optimal balances for specific use cases remains challenging. Systematic evaluation across different source type configurations, holding other variables constant, is needed to map these relationships in a way that can inform practical decisions.

Diversity versus redundancy: Source diversity can improve coverage and user trust. Chen et al. (2022) found users rated responses 41% higher when citing multiple source types. However, diversity also introduces redundancy—multiple sources saying the same thing. The challenge lies in determining when diversity adds value versus merely consuming resources. Asai et al. (2023) tested various combinations and found that a 40-30-30 split (academic-news-technical) optimized their benchmarks, while also acknowledging that optimal ratios likely vary by domain and query distribution.

Credibility and integration complexity: Source type affects perceived credibility independent of factual accuracy. Xiong et al. (2024) found responses citing academic journals received higher trust ratings than general web content, even with identical information. However, credibility varies by audience and domain. Different source types require different processing approaches (citation extraction for papers, date validation for news, and version tracking for documentation), introducing operational complexity that homogeneous collections avoid.

2.2.3 Factor Interaction Issues

Freshness-source type dependencies: Dataset freshness and source type interact in complex ways. Academic papers age more gracefully than news articles in most domains—a five-year-old seminal paper retains value, while last week’s news becomes outdated. However, in rapidly evolving fields like machine learning, even recent academic work may lag practice due to publication delays. Because of these dependencies, freshness requirements differ depending on the type of source. This interaction is one of the reasons a study that examines both factors together, rather than in isolation, is needed.

Domain volatility effects: Domain characteristics moderate how factors affect performance. In high-volatility domains (technology), freshness matters enormously, and source type matters less. In low-volatility domains (history), source type matters more while freshness matters less.

Medium-volatility domains (healthcare) require careful balancing of both factors. Kasai et al. (2023) examined three domains but did not develop frameworks for characterizing volatility in new domains. This research addresses this gap by validating the predictive framework across the three volatility levels.

Non-linear performance relationships: Both factors exhibit non-linear effects complicating optimization. Freshness improvements show diminishing returns—updating daily versus weekly might improve accuracy by 5%, but weekly versus monthly might improve by 15%. Source diversity shows similar patterns. These non-linearities mean simple heuristics like "fresher is always better" prove incorrect. Organizations need frameworks for identifying optimal operating points that consider both factors jointly. Establishing those points through systematic experimentation and modelling is a central aim of this work.

2.2.4 Evaluation and Measurement Challenges

Isolating factor contributions: When RAG systems underperform, diagnosing causes proves difficult. Mallen et al. (2023) found 78% of implementation failures stemmed from dataset problems, but existing evaluation methods struggle to isolate which specific factors caused problems. Standard approaches measure overall performance without attribution. If accuracy drops from 85% to 78%, current methods cannot determine whether this effect reflects freshness decay, source quality degradation, or other causes.

Performance attribution and metric standardization: Even when factor effects are isolated, attributing specific outcomes to factor values remains challenging. Controlled experimentation can address some attribution challenges but requires resources many organizations lack. Additionally, different studies use incompatible metrics (exact match, F1, BLEU, and custom metrics), making cross-study synthesis difficult. Thakur et al. (2021) proposed standardized retrieval benchmarks, but generation quality evaluation remains fragmented. These evaluation

limitations motivate the controlled experimental design this research employs to establish clear factor-performance relationships.

2.3 Traditional RAG Approaches and Performance

Traditional RAG implementations rely on relatively static knowledge bases with periodic updates managed through rule-based heuristics. These approaches lack predictive capabilities, determining factor configurations through trial and error rather than empirical evidence. This section examines conventional approaches that do not attempt performance prediction.

Static knowledge base approaches use knowledge bases that don't change very often, if at all. Lewis et al. (2021) initially demonstrated RAG using Wikipedia snapshots without updates. This minimizes operational complexity but suffers from inevitable staleness. Static approaches work reasonably well only for low-volatility domains where information changes slowly. Performance gradually degrades as information ages, with organizations often discovering problems only when users complain. The lack of proactive maintenance represents a significant weakness directly related to freshness issues (Section 2.2.1).

Periodic update strategies implement updates on fixed schedules—daily, weekly, monthly, or quarterly—typically chosen based on intuition or resource constraints rather than empirical analysis. Periodic strategies improve on static approaches by maintaining some currency but remain suboptimal. Fixed schedules cannot adapt to varying decay rates across content types or domains. Resources may be wasted updating content that remains relevant while other content becomes stale between scheduled updates. Zhang et al. (2022) documented that fixed periodic schedules were often either updated too frequently, wasting resources, or too infrequently, allowing performance degradation.

Rule-based maintenance heuristics rely on practitioner experience rather than systematic evaluation. Common rules include "update technology content weekly, healthcare monthly, and history annually" or "include 60% academic papers, 30% news, and 10% technical docs." These

heuristics offer simplicity and interpretability but lack empirical validation. Rules may reflect conditions in one context that do not generalize to others. Organizations adopting rules from literature or consultants often find they work poorly in their specific situations. The absence of validation makes it difficult to determine when rules apply and when they need modification.

Performance characteristics and limitations: Traditional approaches share several characteristics. Without predictive frameworks, they cannot forecast performance before deployment, leading to costly trial-and-error. They treat factor configuration decisions independently, missing potential interaction effects between freshness and source type. They lack adaptation capabilities—configurations remain fixed even as conditions change. Performance varies widely depending on how well initial configurations match requirements, reflecting the absence of systematic decision frameworks. This is precisely the gap the present study works to fill.

Table 2.1: Comparison of Traditional RAG Approaches

Approach	Update Strategy	Source Selection	Configuration Method	Performance Characteristics	Main Limitations	Representative Studies
Static Knowledge Base	No updates	One-time selection	Domain expertise	High initial, degrades over time	Inevitable staleness; no freshness management	Lewis et al. (2021)
Fixed Periodic Updates	Scheduled intervals	Initial selection, rarely changed	Intuition/constraints	Moderate, varies by schedule	Cannot adapt to domain-specific decay rates	Zhang et al. (2022)
Rule-Based Heuristics	Rule-determined	Rule-based ratios	Literature/consulting	Variable, depends on rule quality	Lacks validation; context-specific performance	Petroni et al. (2021)

Hybrid Periodic-Rule	Combine d scheduling	Rule-based with review	Mixed methods	Better than pure approaches	Still lacks predictive capability	Zhang et al. (2022)
Reactive Maintenance	Problem-triggered	Post-failure adjustment	Trial and error	Poor until tuned	Delayed problem detection; user complaints drive updates	Practitioners
Expert-Curated	Continuous oversight	Expert judgment	Subject matter expertise	High with sufficient resources	Requires scarce expertise; high resource demands	Wang et al. (2022)

2.4 Existing Predictive Systems and Performance

Several research efforts have attempted to develop systems capable of predicting RAG performance. These represent advances over traditional approaches but face significant limitations preventing widespread adoption. This section examines approaches that explicitly attempt performance prediction.

Retrospective analysis systems analyse historical performance data to understand factor-performance relationships. Petroni et al. (2021) developed regression models explaining 32% of performance variance based on knowledge base properties including source diversity, collection size, and domain characteristics. Their approach identified correlations but required substantial historical data for each specific configuration. The retrospective nature limits practical utility—organizations must deploy systems, collect performance data over time, then analyse patterns. This works for understanding past behaviour but does not enable forecasting performance for new configurations before deployment. Developing that prospective capability is one of the motivating contributions of this study.

Statistical prediction models use regression techniques to model factor-performance relationships, attempting to generalize from observed data to predict unseen configurations. Petroni et al. (2021) achieved moderate correlations but explained limited variance, suggesting important factors remained unaccounted for or relationships were more complex than linear models captured. Hofstätter et al. (2023) developed domain-specific statistical models achieving better performance—explaining up to 50% of variance within domains but requiring separate models for each. This technique improves on universal models but increases development overhead. Organizations need separate models for each domain they serve, multiplying resource requirements and limiting practical applicability.

Machine learning prediction approaches treat performance prediction as classification or regression tasks. Kamaloo et al. (2023) trained random forest classifiers, achieving 71% accuracy in predicting whether queries would be answered correctly. Their models used features including document recency, source type distributions, collection size, and domain category. Feature importance analysis validated that dataset-level factors ranked among top predictors—specifically, document age and source compositions supporting the focus of this study on freshness and source type as the primary variables of interest.

However, machine learning approaches require substantial labeled training data—typically thousands of query-performance pairs across various configurations. Organizations rarely have such data initially, making these approaches inaccessible for initial deployment decisions. Empirically validated factor-performance relationships, established through controlled experimentation, offer a more accessible starting point for organizations that do not yet have their own historical data.

Temporal decay modeling represents another prediction approach. Metzler et al. (2021) developed decay functions modeling how relevance decreases over time—news articles received aggressive decay (30-day half-life), while academic papers decayed slowly (2-year half-life). Applying these weights during retrieval improved temporal accuracy by 18% on time-sensitive

queries. However, approaches require domain-specific decay parameter tuning—parameters optimized for news do not transfer to other domains. This limitation reinforces the need to examine freshness impact across domains with different volatility characteristics, establishing relationships that are more generalizable across deployment contexts.

Performance and limitations: Predictive systems show promise but face four key limitations. First, most require extensive training data covering specific configurations being predicted organizations implementing RAG for the first time lack this data. Second, models often do not generalize across domains, requiring retraining for each new application. Third, predictions typically focus on overall accuracy without decomposing contributions from specific factors such as dataset freshness and source type. Fourth, most systems can predict performance for familiar configuration types but struggle with new combinations. Addressing these limitations requires controlled experimentation that deliberately varies both factors across multiple freshness levels and source type configurations, enabling benchmark performance ranges to be established for a range of practical scenarios.

Table 2.2: Comparison of Predictive RAG Systems

System Type	Prediction Method	Data Requirements	Generalizability	Accuracy	Factor Isolation	Key Limitations	Representative Studies
Retrospective Regression	Statistical modeling of historical data	Extensive historical data	Domain-specific	Moderate (R ² =0.32)	Limited	Requires historical data; cannot predict new configurations	Petroni et al. (2021)
Domain-Specific Statistical	Regression models per domain	Moderate per domain	Single domain	Better (R ² =0.50)	Limited	Separate models needed per domain	Hofstätter et al. (2023)
ML Classification	Random forests, boosting	Large labeled training set	Moderate with retraining	Good (71% accuracy)	Moderate	Data-hungry; binary predictions only	Kamalloo et al. (2023)

ML Regression	Continuous prediction	Large labeled training set	Moderate with retraining	Moderate	Moderate	Struggles with exact performance scores	Kamalloo et al. (2023)
Temporal Decay Models	Mathematical decay functions	Moderate longitudinal data	Domain-specific	Good for temporal queries (+18%)	High for freshness	Requires decay parameter tuning per domain	Metzler et al. (2021)
Hybrid Ensemble	Combines approaches	Moderate to high	Better than individual	Best (81% domain-specific)	Limited	Complex implementation; high overhead	Hofstätter et al. (2023)

2.5 Existing Solutions for Factor Optimization

Beyond predictive systems, researchers have developed various solutions specifically targeting dataset-level factor optimization. These solutions focus on practical methods for managing freshness and source selection, the two dataset characteristics at the center of this study.

Freshness management solutions address temporal decay through several approaches. Selective update strategies target volatile content for frequent updates while leaving stable content unchanged. Zhang et al. (2022) documented that organizations focusing updates on the most volatile content achieved near full-update performance at substantially lower resource cost. Implementation requires identifying volatile content through metadata (publication dates, content categories) or performance monitoring to detect degradation. These findings are directly relevant to understanding how dataset age shapes system performance and quantifying that relationship under controlled conditions is one of the goals of the present study.

Incremental indexing reduces update costs by adding new documents without reindexing existing content, maintaining reasonable freshness while consuming fewer resources. Gao et al. (2023) found incremental updates required 10-20% of full reindexing computational costs. However,

incremental approaches can introduce inconsistencies as vector spaces drift, necessitating periodic full reindexing to maintain quality.

Source selection and curation solutions address source type optimization. Quality scoring frameworks quantify source quality through metrics including author authority (citation counts, h-index), publication venue prestige (journal impact factors), content characteristics (writing quality, citation density), and community signals (upvotes, shares). Wang et al. (2022) developed composite quality scores incorporating multiple signals. Quality scoring enables filtering low-quality sources and prioritizing high-quality ones, though quality metrics often correlate with publication delay—the highest-quality sources (peer-reviewed journals) lag current events most significantly. This creates a tradeoff between quality and timeliness that an integrated framework examining both factors together must account for.

Diversity optimization approaches include maximizing topic coverage, balancing source types, and detecting redundancy. Asai et al. (2023) developed algorithms optimizing diversity subject to budget constraints. These approaches help but require defining optimization objectives a priori—what constitutes appropriate diversity depends on anticipated query distributions and user preferences, factors varying across applications. Systematically evaluating how different source type configurations affect performance across multiple metrics provides the kind of evidence needed to make these design choices on firmer footing.

Automated curation pipelines reduce human labor by identifying candidate sources, assessing quality, classifying content types, and recommending inclusion/exclusion. Chen et al. (2022) developed pipelines reducing manual curation effort by 60% while maintaining comparable selection quality. However, automation cannot fully replace human judgment for edge cases, context-dependent quality assessments, and strategic sourcing decisions.

Hybrid and adaptive solutions combine multiple optimization strategies. Kasai et al. (2023) developed hybrid systems that adjusted factor configurations based on detected performance patterns. Adaptive systems attempt to learn optimal configurations through reinforcement learning

or evolutionary approaches, exploring alternatives and converging toward good solutions. However, exploration periods may degrade user experience, limiting practical applicability.

Comparative analysis and limitations: Existing solutions show promise but face integration challenges. Most optimize single factors in isolation rather than considering how freshness and source type interact. Freshness management systems assume source selection is fixed. Source curation systems assume update schedules are predetermined separately. Coordinating these decisions remains an open problem. Additionally, most solutions require domain-specific tuning—parameters optimized for one context rarely transfer to others. Organizations need extensive experimentation to identify appropriate parameters or frameworks that predict good starting points based on domain characteristics. Providing that kind of validated, cross-domain guidance is one of the distinguishing contributions this study aims to make.

Table 2.3: Comparison of Factor-Based Solutions

Solution Category	Primary Factor Focus	Key Techniques	Performance Improvement	Resource Efficiency	Generalizability	Main Limitations	Representative Studies
Selective Updates	Freshness	Content volatility classification	90% of full-update quality	60% resource reduction	Good with proper detection	Requires volatility classification mechanism	Zhang et al. (2022)
Temporal Decay Weighting	Freshness	Mathematical decay functions	+18% temporal accuracy	High (minimal overhead)	Domain-specific	Needs decay calibration per domain	Metzler et al. (2021)
Incremental Indexing	Freshness	Append-only indexing	Near full-update quality	80-90% cost reduction	Good for frequent updates	Can introduce inconsistencies over time	Gao et al. (2023)

Quality Scoring	Source type	Multi-signal quality metrics	+27% with high-quality sources	N/A	Moderate	Quality-timeliness tradeoffs; context-dependent	Wang et al. (2022)
Diversity Optimization	Source type	Topic/type balancing	Variable by objective	N/A	Requires objectives	Needs predefined optimization objectives	Asai et al. (2023)
Automated Curation	Source type	ML-based assessment	Comparable to manual	60% effort reduction	Moderate	Cannot fully replace human judgment	Chen et al. (2022)
Hybrid Factor Management	Both factors	Combined techniques	Better than single-factor	Variable	Limited	Integration complexity; lacks unified framework	Kasai et al. (2023)

2.6 Evaluation Methods for RAG Systems

Measuring RAG performance requires methodologies that assess both retrieval and generation quality while enabling attribution of outcomes to specific factors. This section examines evaluation approaches relevant to the controlled experimentation and framework validation that this study employs.

Retrieval quality metrics adapt traditional information retrieval measures to RAG contexts. Precision at k ($P@k$) measures the proportion of top- k retrieved documents that are relevant. Thakur et al. (2021) used $P@10$ across BEIR benchmark datasets, finding substantial variance across domains (0.42 to 0.78). However, relevance judgments that prove challenging documents may be topically relevant but temporally outdated, or from source types that are structurally mismatched to the query. Mean Average Precision (MAP) summarizes precision across multiple recall levels, providing a single quality score. Normalized Discounted Cumulative Gain (nDCG) accounts for ranking quality—relevant documents appearing earlier contribute more than those

later. Ram et al. (2023) found $nDCG@10$ correlated strongly with generation quality, suggesting retrieval ranking matters significantly.

These metrics assess whether relevant documents are retrieved but do not measure factor-specific properties. Evaluating the effect of dataset freshness and source type specifically requires going beyond overall retrieval scores. It requires systematic manipulation of these characteristics under controlled conditions, with direct comparisons across freshness levels and source type configurations.

Generation quality metrics prove more challenging. Reference-based metrics compare generated text against reference answers. BLEU measures n-gram overlap, ROUGE focuses on recall, and METEOR incorporates synonyms. These work reasonably well for factual questions with clear correct answers but struggle with open-ended queries. Factual accuracy assesses whether generated claims match ground truth. Honovich et al. (2022) developed TRUE, which decomposes responses into atomic claims and verifies each against retrieved documents using NLI entailment checking, achieving 84% accuracy in identifying unsupported claims across summarization and dialogue tasks.

Hallucination detection identifies generated content not supported by retrieved documents. Honovich et al. (2022) found 28% of hallucinations originated from retrieved content rather than generation errors, highlighting dataset quality importance. This motivates research to focus on dataset-level factors (freshness and source type) rather than sole generation improvements. Detection methods include entailment checking (verify claims are entailed by documents), factuality scoring (trained classifiers identifying likely hallucinations), and self-consistency checks. Factor-specific evaluation approaches enable isolating contributions from freshness and source type. Temporal evaluation measures how performance changes as knowledge bases age. Zhang et al. (2022) tracked accuracy monthly over six months, documenting 2.3% monthly decline in technology domains. This approach directly assesses freshness effects but requires longitudinal data collection. Cross-sectional approaches using documents of varying ages offer a more

efficient alternative for examining how dataset age influences performance across different domain contexts.

Source attribution analysis examines which sources contribute to responses and how source types affect outcomes. Chen et al. (2022) tracked which documents influenced generation through attention weights, finding 60% of generated content came from top-3 retrieved documents. Source type analysis revealed that official documentation dominated technical queries while community forums influenced troubleshooting. Extending such analysis through systematic source type manipulation—comparing performance across single-source and mixed configurations—allows clearer conclusions about how source composition shapes generation quality.

Ablation studies systematically remove or modify factors to measure individual contributions. By comparing performance with and without specific source types, or with different freshness levels, ablation studies isolate factor effects. Ram et al. (2023) used ablation to decompose retrieval versus generation contributions. Employing similar systematic factor manipulation across controlled conditions enables clear attribution of performance outcomes to dataset freshness, source type, and the interaction between the two.

Integrated evaluation and limitations: Comprehensive evaluation requires integrating multiple metrics. KILT (Knowledge Intensive Language Tasks) provides standardized benchmarks for knowledge-grounded generation but focuses primarily on accuracy without systematically evaluating factor effects. BEIR (Benchmarking IR) offers diverse retrieval evaluation across 18 datasets spanning domains, demonstrating that models optimized for one domain often performed poorly on others. However, BEIR evaluates only retrieval, not end-to-end RAG performance.

Several important gaps in current evaluation approaches connect directly to the problems identified in Chapter 1. First, most methods assess overall system performance without isolating the contribution of individual dataset-level factors. When accuracy drops, it remains unclear whether the cause is freshness decay, poor source quality, or their interaction—precisely the

difficulty described in Section 1.2.5. Second, existing frameworks evaluate factors independently, leaving the interaction effects between freshness and source type largely unmeasured (Section 1.2.4). Third, evaluation is predominantly retrospective: methods examine performance after deployment rather than offering prospective estimates that would help organizations plan knowledge base configurations in advance. This absence of forward-looking evaluation is a direct expression of the resource allocation uncertainty described in Section 1.2.1. Fourth, no standardized methodology connects freshness and source type metrics to measurable performance outcomes in a way that generalizes across domains, reinforcing the lack of predictive frameworks identified in Section 1.2.7.

This study addresses these gaps through controlled experimentation that systematically varies dataset freshness across four levels and source type across three configurations, within three domains selected for their different volatility characteristics. This design enables clear factor attribution, interaction assessment, and empirical relationship establishment—supporting both prospective performance prediction and the development of benchmark guidelines for practitioners.

2.7 Research Gaps and Framework Positioning

Synthesizing findings across resource allocation, implementation issues, existing approaches, and evaluation methods reveals systematic gaps that this research addresses through its predictive framework based on relevant dataset-level factors.

Gap 1: Absence of prospective predictive frameworks - As documented in Tables 2.1-2.3, existing approaches are predominantly retrospective. Traditional implementations (Table 2.1) rely on trial-and-error or unvalidated rules. Predictive systems (Table 2.2) require extensive historical data for specific configurations, making them inaccessible for initial deployment decisions. Factor optimization solutions (Table 2.3) address individual factors without integrated prediction across multiple factors. Organizations need prospective frameworks forecasting performance for proposed configurations before deployment. This study develops such a capability by building

and validating an integrated framework that translates dataset characteristics into performance estimates, enabling evidence-based decisions about factor configurations without costly experimentation.

Gap 2: Limited factor integration—A review of critical issues (Section 2.2) revealed that freshness and source type interact in complex, domain-dependent ways. However, existing solutions (Table 2.3) predominantly optimize factors independently. Freshness management assumes source selection is fixed. Source curation assumes update schedules are predetermined separately. No systematic frameworks integrate multiple factors while accounting for interactions. This fragmentation stems partly from evaluation limitations—current methods struggle to assess how factors jointly affect performance. This study addresses the gap through systematic examination of both dataset freshness and source type, employing controlled experimentation that isolates and combines their effects to establish how they interact across different domain contexts.

Gap 3: Insufficient generalizability—Examination of predictive systems (Table 2.2) and solutions (Table 2.3) shows most approaches require domain-specific calibration or extensive tuning. Statistical models achieve reasonable performance within domains but require separate models for each application. Machine learning approaches need large, labeled datasets for each specific context. Organizations implementing RAG for the first time lack domain-specific data for calibration. They need frameworks that generalize reasonably across domains with minimal customization or principled methods for rapid adaptation. This study establishes generalizability by validating the predictive framework across three domains with different volatility characteristics, directly addressing the domain-specific variations identified in Section 2.2.1 and documenting the boundary conditions within which predictions can be trusted.

Gap 4: Resource-performance tradeoff neglect - Resource allocation analysis (Section 2.1) highlighted that organizations face real constraints—limited budgets, personnel, and computational resources. However, existing solutions rarely consider efficiency explicitly. Few

approaches optimize resource-performance tradeoffs; most focus solely on maximizing performance regardless of cost. The field needs frameworks enabling identification of configurations providing acceptable performance within resource constraints. Organizations should be able to determine how much performance they sacrifice by updating monthly instead of weekly and whether that tradeoff suits their constraints. This study addresses this gap by establishing performance benchmarks for different combinations of freshness and source type, characterizing how accuracy changes with more frequent updates or more diverse source configurations, and thereby informing resource allocation decisions with quantitative evidence.

Novelty and contributions - This study is among the first to develop and validate a prospective predictive framework for RAG performance based on systematically quantified dataset-level characteristics—specifically, freshness and source type as representative and controllable factors—across domains with different volatility profiles. The framework enables: (1) Prospective prediction—estimating performance for untested configurations before deployment, rather than requiring historical data for each specific combination; (2) Integrated multi-factor analysis—accounting for freshness-source type interactions rather than treating the two variables independently; (3) Cross-domain validation—establishing generalizable relationships and their boundary conditions across high-, medium-, and low-volatility domains; (4) Resource-aware guidance—providing concrete performance benchmarks that allow organizations to evaluate tradeoffs and select configurations suited to their operational constraints.

The framework illustrates a broader principle: dataset-level factors can be systematically measured and incorporated into predictive models. While the study demonstrates this through two factors, the methodology establishes a replicable approach that could be extended to additional dataset properties in future work. The contribution is not a comprehensive solution to all RAG performance challenges, but a principled, empirically grounded starting point for factor-based decision-making.

CHAPTER 3: RESEARCH METHODOLOGY

3.1 Overview

Designing a methodology for this study required working through a set of interrelated decisions that are not entirely independent of each other. The central question—whether dataset freshness and source type can predict RAG system performance—demands conditions where those two properties genuinely vary while everything else stays the same. That requirement shapes almost every subsequent choice, from how documents are collected to how statistical results are interpreted.

The chapter is organized to follow the phases set out in Table 3.1, the research design summary, which maps each phase directly to a research objective and research question. Rather than presenting methodology as a list of procedures, the account below explains the reasoning behind choices. Why these three domains? Why these freshness windows? Why hold the retrieval model constant rather than testing multiple? These are decisions made for reasons that a reader should be able to evaluate.

The study takes a positivist, quantitative stance. Performance outcomes of RAG systems—retrieval precision, response accuracy, and hallucination rate—are measurable quantities. The relationships being investigated are causal in intent: the study tests whether varying one thing (dataset characteristics) produces consistent changes in another (system performance). A deductive design, where hypotheses are drawn from existing literature and tested against controlled experimental data, is appropriate for that kind of inquiry (Creswell and Creswell, 2018). Before proceeding to the phases, one term requires a precise definition. In this study, a predictive framework refers to a structured, factor-based system that quantifies dataset-level characteristics, models their interactions, and translates the resulting relationships into performance estimation ranges and deployment guidelines. Statistical and machine learning methods are used within the framework as analytical and validation mechanisms; they do not constitute the framework itself.

The framework predicts expected performance ranges across three dimensions: expected retrieval precision range, expected response accuracy range, and expected hallucination rate range, given a particular combination of dataset freshness and source type. It does not predict exact response content, individual query outcomes, or model behavior at the token level. This distinction is deliberate—the framework is designed to support decisions about knowledge base composition and update strategy before a system is deployed, not to simulate the system itself. Table 3.1 sets out the methodology according to the study’s four research objectives. Each row begins with a research objective, identifies the research question it directly addresses, describes the methods used to pursue that objective, and states the expected output. This structure ensures that the connection between purpose and method is made explicit at each stage of the study, rather than presenting methodology as a disconnected sequence of procedures.

Table 3.1: Research Design Summary — Research Objectives, Research Questions, Methods, and Expected Outputs

Phase	RO	Research Question Addressed	Method / Activity	Expected Output
Phase 1 <i>Factor-Performance Analysis</i>	RO1 <i>Analyze key factors (dataset freshness and source type) and their relationship with RAG system performance across high-, medium-, and low-volatility domains.</i>	RQ1 How do dataset freshness and source type, as key dataset-level factors, relate to RAG system performance across high-, medium-, and low-volatility domains, measured as changes in retrieval precision, response accuracy, and hallucination rate?	• Construct three domain datasets (technology, healthcare, historical research) across four freshness levels and three source type configurations • Assign continuous freshness scores using document metadata • Develop 200-query bank per condition with ground-truth answers; build FAISS retrieval index • Collect primary metrics (P@5, nDCG@5, BERTScore, hallucination rate via DeBERTa NLI) holding DPR and T5-large constant • Run two-way ANOVA with post-hoc Tukey HSD (Bonferroni-corrected); report effect sizes (η^2)	• Nine annotated document collections with verified freshness scores • Validated 2,400-query bank with ground-truth answers and FAISS indices per condition • Quantified freshness–performance relationships across three domain volatility levels (ANOVA results with effect sizes and post-hoc comparisons)
Phase 2 <i>Experimental Design and Execution</i>	RO2 <i>Examine the impact of source type, as a relevant factor, on</i>	RQ2 What is the individual contribution of	• Run controlled experiments across twelve conditions; pose identical queries to all three source configurations per domain	• Performance profiles per source type configuration across all five metrics

Phase	RO	Research Question Addressed	Method / Activity	Expected Output
	<i>the performance of RAG across multiple metrics.</i>	source type configurations to RAG system performance across multiple evaluation metrics, and does this contribution vary by domain volatility?	• Collect automated metrics (BERTScore, ROUGE-L, METEOR, DeBERTa NLI) and human expert ratings (5-point Likert, $\kappa \geq 0.70$, stratified 20% sample) • Compute Source Diversity Index (Shannon entropy) as continuous variable • Apply two-way ANOVA across all three domain contexts to test whether source type effects are stable or domain-dependent	• Computed Source Diversity Index values linked to performance outcomes • ANOVA results and effect sizes showing which source configurations produce gains and which introduce redundancy • Human evaluation scores with confirmed inter-rater reliability ($\kappa \geq 0.70$)
Phase 3 <i>Framework Development</i>	RO3 <i>Develop a predictive framework capable of estimating RAG effectiveness based on relevant factors such as dataset freshness and source type.</i>	RQ3 Can a predictive framework be developed that estimates RAG system performance from dataset-level factors, specifically freshness and source type, across domains with different volatility characteristics?	• Apply multiple regression (OLS with freshness score, Source Diversity Index, domain volatility, and interaction terms) to model factor–performance relationships • Run Random Forest analysis (500 trees, grid-searched hyperparameters) as non-parametric validation layer • Apply 10-fold cross-validation stratified by domain and source-type condition • Fit exponential and polynomial decay models to the freshness dimension per domain volatility level	• Cross-validated regression equations and Random Forest feature importance rankings • Decay curves parameterised by domain volatility • Predictive framework with factor-based estimates of expected retrieval precision, response accuracy, and hallucination rate (with cross-validation error bounds)
Phase 4 <i>Framework Validation and Benchmarking</i>	RO4 <i>Evaluate the proposed predictive framework using benchmark performance ranges established across different factor configurations and domain contexts.</i>	RQ4 How accurately does the proposed predictive framework estimate RAG performance across different factor configurations and domain contexts, and what are its reliable operating boundaries?	• Evaluate the RO3 framework against a held-out set of experimental conditions not used in model fitting • Establish benchmark performance ranges (95% CI) per factor configuration and domain context • Document boundary conditions where framework estimates are reliable and where they are not	• Validated predictive framework with benchmark performance ranges (95% CI) per domain and factor configuration • Documented boundary conditions stating where estimates are reliable and where they are not • Deployment guidelines enabling evidence-based decisions about knowledge base composition and update frequency

3.2 Research Design

The study uses a quasi-experimental design. Full randomization is not possible here because freshness is an inherent property of documents, not an assignable treatment. 2021 published in

2021 cannot be experimentally assigned to the "one week old" condition. Conditions are therefore constructed by selecting documents from specific temporal windows, and comparisons are made between those conditions while everything else is kept constant. This is a standard design choice in temporal information retrieval research, and the inferential risk it introduces—that documents in different time windows might differ in ways other than age—is addressed by controlling collection procedures carefully (Shadish, Cook, and Campbell, 2002).

Two factors are manipulated: dataset freshness and source type. Freshness is operationalized at four levels: documents are less than one week old, one week to one month, one to six months, and older than six months. Source type is manipulated at three levels: a single-source configuration using academic papers only, a mixed configuration adding news articles, and a diverse configuration that also includes technical documentation. These two factors are crossed within three domain contexts—technology, healthcare, and historical research—producing twelve experimental conditions in total.

The study employs a domain-adjusted factorial design rather than a full $4 \times 3 \times 3$ cross because certain freshness windows are not meaningful in low-volatility domains. A "less than one week old" condition has no practical equivalent in historical research, where the relevant timescale is years rather than days. Freshness windows are therefore adjusted at the domain level to preserve conceptual equivalence across domains while reflecting genuinely different rates of information change. To ensure that this domain adjustment does not compromise comparability across domains, freshness is analyzed both categorically—using the four windows for condition-level comparison—and as a continuous variable measured in days since publication, which can be directly compared across domains in regression modelling regardless of the absolute time values involved.

Domain context is not an independent variable in the manipulative sense. Technology, healthcare, and historical research were selected because they represent genuinely different rates of information change. Technology is high-volatility; information that was accurate last month may

be outdated today. Healthcare is medium-volatility; clinical evidence evolves steadily but clinical practice changes more slowly. Historical research is low-volatility; the factual record is stable even if scholarly interpretation shifts. These three contexts allow the study to test whether freshness and source type effects generalize or whether they are specific to a particular kind of domain.

One design decision deserves particular attention. The retrieval model, the generation model, and the prompt template are all held constant across every experimental condition. This is deliberate. If the architecture were allowed to vary, any observed performance differences would be confounded—it would be impossible to tell whether a change in retrieval precision reflected a dataset characteristic or a model choice. The trade-off is that findings apply most directly to the specific architecture used. The expectation, grounded in prior work, is that factor-performance relationships are sufficiently general to hold across architectures, since freshness and source type are properties of documents rather than of models.

The methodology follows four sequential phases, each corresponding to one of the research objectives. Phase 1 (addressing RO1) constructs the document collections and establishes how dataset freshness and source type relate to RAG performance across the three domain volatility levels. Phase 2 (addressing RO2) runs the controlled experiments across the twelve conditions to examine how source type configurations contribute to performance across multiple metrics. Phase 3 (addressing RO3) applies statistical and machine learning analysis to build the predictive framework. Phase 4 (addressing RO4) validates the framework against held-out conditions and establishes benchmark performance ranges with documented boundary conditions. Each phase produces outputs that feed directly into the next.

The process allows targeted iteration. If Phase 4 validation reveals that the framework's predictions diverge substantially from observed performance in certain conditions, additional experiments may be warranted before finalizing the framework's coverage claims. This iteration possibility is built into the design rather than treated as an exception.

3.3 Phase 1: Factor-Performance Analysis

3.3.1 Dataset Construction

Domain selection was guided by one requirement above others: the domains needed to differ in how quickly their information becomes outdated. If all three domains aged at the same rate, the study would have no way to test whether freshness effects are domain-dependent—a question that matters directly for the framework's practical scope.

Technology was selected because it is consistently characterized in prior literature as high-volatility. Zhang et al. (2022) documented a 2.3% monthly decline in retrieval accuracy for technology-related queries when knowledge bases were not updated. That figure is specific enough to be useful as a comparison point for this study's own freshness measurements. Healthcare sits in a middle position. Clinical evidence accumulates continuously through trials and meta-analyses, but treatment protocols update quarterly or annually rather than daily. This creates a domain where freshness matters but is not the dominant factor it becomes in technology. Historical research changes more slowly than either. The primary factual record is stable, though scholarly consensus shifts gradually. Its inclusion in the study establishes the low end of the volatility spectrum and provides a baseline condition for the framework.

Three domains is also a practical constraint. Adding a fourth or fifth domain would increase data collection, expert validation, and analysis work proportionally. Three provides enough variation to test whether the framework's factor-performance relationships are domain-general without making the study unmanageable.

3.3.2 Source Type Classification

Three source type categories are used: academic publications, news and periodical articles, and technical documentation. They represent structurally and functionally distinct document types that appear consistently across all three domains. Academic papers are peer-reviewed, citation-dense, and typically lag current events due to publication timelines. News articles are timely, varied in editorial quality, and structured for rapid consumption. Technical documentation—

software manuals, clinical protocols, official API references—is procedural, precise within its stated scope, and updated in synchrony with the systems it describes.

Each combination of domain and source type produces a distinct sub-collection. A technology news article and a healthcare news article are both 'news,' but they differ in editorial process, typical update cycle, and the nature of claims they make. The framework needs to account for this. Source type classification is therefore treated as a domain-relative variable in the analysis, not as a domain-independent label.

Document classification uses a combination of metadata field publisher type, presence or absence of peer-review indicators, structural features such as citation sections, version numbers, and publication venue. Documents that cannot be classified unambiguously are excluded. The goal is not an exhaustive or perfectly clean dataset, but one where the source type conditions can be trusted to reflect genuine structural differences rather than misclassification noise.

3.3.3 Freshness Definition and Windows

Freshness is operationalized as elapsed time between a document's publication date and the date on which it is retrieved during the experiment. This is computed as a continuous measure—days since publication—and then assigned to one of four categorical windows for the purpose of condition construction: less than one week, one week to one month, one to six months, and older than six months.

The four windows were chosen to span the range of information ages that organizations typically encounter in RAG deployments, from near-real-time retrieval to knowledge bases that have not been refreshed in half a year. Zhang et al. (2022) tracked performance across a six-month window; Metzler et al. (2021) used windows of similar granularity when calibrating temporal decay functions. The windows used here are consistent with that precedent and are wide enough to contain sufficient documents in each domain.

For the historical domain, the same categorical structure is used but shifted upward in absolute terms. Historical documents are measured in years rather than weeks, because the volatility of that domain is low enough that a three-year-old archive document is functionally comparable to a one-month-old technology document in terms of expected accuracy. Domain-adjusted freshness windows prevent the freshness variable from becoming trivially skewed in low-volatility conditions.

3.3.4 Document Sources, Collection Scope, and Quality Assurance

Table 3.2 — presented later in Section 3.4.2 — defines the twelve experimental conditions, but the document collections that populate those conditions are described here. Table 3.4 below lists each domain-source type combination, the specific repositories or outlets used, the date range of collected documents, their typical update frequency, and the reasoning behind their inclusion. Including a 'reason for inclusion' column in the table is deliberate. Source choices are not self-evident, and the reasoning behind them matters for assessing whether the collection represents the kind of real-world knowledge base an organization might maintain. Each source is present because it has a distinct role in its domain-source type combination, not merely because it was convenient to access.

Note: Table 3.1 (Research Design Summary) sets out the method for RO1 at a design level. Table 3.4 below provides the operational specifics — the exact sources, date ranges, update frequencies, and inclusion rationale — needed to implement the dataset construction activity in practice.

Table 3.4: Document Collections by Domain, Source Type, and Inclusion Rationale

Domain	Source Category	Specific Sources	Date Range Collected	Update Frequency	Reason for Inclusion
Technology (High Volatility)	Academic	IEEE Xplore, arXiv cs.IR and cs.AI, ACL Anthology papers	2020 – 2025	Daily to weekly	Peer-reviewed technical literature captures methodological developments, but publication delays mean it may lag practical developments by months.

Technology (High Volatility)	News	TechCrunch, MIT Technology Review, Wired, The Verge	2020 – 2025	Daily	News sources provide temporal currency that academic literature cannot. They cover product releases, regulatory changes, and community discussions in near real-time.
Technology (High Volatility)	Technical Documentation	GitHub Docs, official API documentation, Stack Overflow accepted answers	2020 – 2025	Weekly	Technical documentation reflects what practitioners actually use. It is often the most actionable source for implementation queries and updates with software releases.
Healthcare (Medium Volatility)	Academic	PubMed Central, NEJM, The Lancet, Cochrane systematic reviews	2018 – 2025	Monthly	Clinical evidence accumulates steadily rather than in bursts. Journal articles and systematic reviews represent the standard of evidence that clinical decisions rely on.
Healthcare (Medium Volatility)	News	WHO situation reports, MedPage Today, CDC health bulletins	2019 – 2025	Monthly to quarterly	Health news provides updates on emerging diseases, drug approvals, and guideline changes that journals document later. These are particularly relevant for time-sensitive health queries.
Healthcare (Medium Volatility)	Technical Documentation	NHS clinical protocols, CDC treatment guidelines, drug prescribing information sheets	2018 – 2025	Quarterly	Clinical protocols and prescribing guides are structured, authoritative, and regularly revised. They represent a distinct document type with high factual reliability for specific clinical queries.
Historical Research (Low Volatility)	Academic	JSTOR, Project MUSE, Cambridge Histories Online, archive journals	2015 – 2025	Annual	Historical scholarship changes slowly. Peer-reviewed articles and edited volumes form the principal evidence base for historical queries, and their factual core remains stable over years.

Historical Research (Low Volatility)	Archival / Primary	Europeana digital archive, Library of Congress digital collections, National Archives digitised records	2015 – 2025	Annual	Primary sources provide direct evidence rather than interpretation. Including them tests whether source type diversity—factual documents versus interpretive scholarship—affects retrieval performance even in a low-volatility domain.
Historical Research (Low Volatility)	Technical Documentation	Museum catalogue records, digitised reference encyclopaedias, atlas entries	2015 – 2025	Annual	Reference texts and catalogue entries occupy a distinct structural category from narrative scholarship. Their inclusion creates genuine source-type diversity in the historical domain collection.

Data quality is assured through four steps applied to each collection. First, sources are verified against their respective publisher or platform databases to confirm that collected documents are authentic and complete. Second, near-duplicate detection is run using cosine similarity at a threshold of 0.95—documents above this threshold are treated as duplicates and one copy is removed. Third, a 10% random sample from each collection is reviewed by a domain expert who confirms that documents are correctly classified by source type and domain, and that metadata timestamps are accurate. Fourth, all metadata is standardized according to a schema defined before collection begins, covering publication date, source type classification, domain, and publisher identifier. Documents that fail any of these checks are removed and replaced from the same source and date range.

The target is a dataset where quality differences between conditions can be attributed to freshness level and source type, not to collection inconsistencies. Perfect cleanliness is not the goal—real RAG deployments use imperfect data—but systematic noise that would obscure factor effects must be avoided.

3.4 Phase 2: Experimental Design and Execution

The query bank is constructed before any experiments are run. Constructing queries after seeing experimental results would introduce selection bias—an unconscious tendency to favour queries that produce interpretable patterns. A pre-constructed bank fixes the evaluation targets in advance.

Queries are organized into three types. Factual queries have a single verifiable correct answer that can be confirmed against a specific document in the collection, for example, a question about a drug's approved dosage or the publication year of a specific technical standard. Analytical queries require integrating information from multiple retrieved passages to produce a coherent response, for example, summarizing the current state of evidence on a clinical intervention. Comparative queries ask the system to contrast two things: two time periods, two methods, two institutions. Including all three types ensures that the study captures freshness and source type effects across different cognitive demands, not just the easiest factual retrieval case.

Approximately 20% of queries in each domain are designed to be time sensitive. A query is classified as time-sensitive only if the ground-truth answer changes across at least two freshness windows within that domain—that is, the correct answer using documents less than one month old differs from the correct answer using documents older than six months. This definition ensures that time-sensitivity is a verified property of the query, not a subjective label. The remaining 80% are time-neutral: the correct answer does not change across the temporal range of the dataset. Time-neutral queries provide a baseline and allow the study to isolate freshness effects by comparing time-sensitive and time-neutral performance separately.

Within each domain, the same query bank is used across all source-type configurations. This means that performance differences between the academic-only, mixed, and diverse source conditions reflect differences in what is available for retrieval—not differences in what is being asked. Holding query content constant across source-type conditions is a straightforward internal

validity measure, but it is worth stating explicitly because it is the mechanism that makes source-type comparisons interpretable.

The sample size of 200 queries per experimental condition was determined by power analysis. For a two-way ANOVA aiming to detect a medium effect size (Cohen's $f = 0.25$) at $\alpha = 0.05$ with power of 0.80, a minimum of 128 observations per cell is required. Using 200 per condition provides a reasonable buffer for subgroup analysis and any data exclusions arising from quality checks. In practice, final query counts may range between 150 and 200 per condition depending on pilot validation outcomes—queries that fail expert review or that prove unanswerable within the collected documents will be replaced or excluded before the main experiment begins.

Expert validation is the final step before the bank is used. Domain experts confirm that each query has a clear, verifiable answer within the document collections, that difficulty levels are realistic, and that reference answers are accurate. Queries where the correct answer can be retrieved from a model's parametric knowledge alone—without consulting any document—are flagged and excluded from the primary analysis. If not excluded, such queries would artificially inflate performance scores in conditions where documents provide little relevant information.

3.4.1 Experimental Conditions and System Configuration

3.4.2 The Twelve Experimental Conditions

The twelve experimental conditions follow directly from a factorial design structure: four freshness levels crossed with three source type configurations produces $4 \times 3 = 12$ distinct combinations. This approach is consistent with factorial experimental design principles in information retrieval research, where crossing independent variables allows systematic examination of both main effects and interaction effects (Shadish, Cook, and Campbell, 2002). The four freshness levels (≤ 1 week, 1 week–1 month, 1–6 months, and >6 months) reflect the decay intervals documented by Zhang et al. (2022) and the volatility-based staging used in RAG evaluation benchmarks

(Thakur et al., 2021). The three source type configurations (academic only; academic and news; academic, news, and technical documentation) represent structurally distinct document categories identified in the literature as having measurably different effects on retrieval and generation quality (Ram et al., 2023). Not every crossing is equally straightforward. The historical domain cannot produce a ‘less than one week old’ condition because the nature of historical scholarship means no documents meet that criterion. Conditions are therefore adjusted at the domain level—historical conditions use longer freshness windows that are contextually equivalent given the domain’s lower volatility. Table 3.2 lists all twelve conditions and includes a rationale column explaining what each condition is designed to test. Reading across the table makes the logic of the design visible: each condition either isolates one variable, tests a specific interaction, or establishes a domain-level baseline.

The twelve conditions below were derived from the literature reviewed in Chapter 2. Freshness windows follow decay patterns documented by Zhang et al. (2022) and volatility profiles from Izacard et al. (2023). The three source-type configurations reflect structurally distinct document categories identified by Gao et al. (2023) as having measurably different effects on retrieval precision, factual accuracy, and contemporary relevance. These conditions are cross-referenced in Table 3.1, where they are mapped to RO1 and RO2.

Table 3.2: Twelve Experimental Conditions — Freshness, Source Type, Domain, and Condition Rationale

No.	Domain	Volatility	Freshness Window	Source Configuration	Rationale for Condition
C1	Technology	High	≤ 1 week	Academic only	Tests on whether very recent academic content alone sustains retrieval precision in fast-moving domains, based on freshness decay findings (Zhang et al., 2022).

C2	Technology	High	≤ 1 week	Academic + News	Tests whether adding news improves temporal coverage, motivated by source type complementarity (Gao et al., 2023)..
C3	Technology	High	≤ 1 week	Academic + News + Technical documentation	Tests whether maximal source diversity improves performance or introduces noise, based on diversity vs redundancy trade-offs (Chen et al., 2022).
C4	Technology	High	1 week to 1 month	Academic only	Measures short-term freshness decay effect in high-volatility domain (Zhang et al., 2022).
C5	Technology	High	1 week to 1 month	Academic + News	Tests interaction between moderate freshness and mixed sources.
C6	Technology	High	1 week to 1 month	Academic + News + Technical documentation	Tests whether diversity helps or hurts under moderate staleness conditions.
C7	Healthcare	Medium	1–6 months	Academic only	Tests performance in medium-volatility domain where evidence evolves steadily (Kasai et al., 2023).
C8	Healthcare	Medium	1–6 months	Academic + News	Tests whether timely health reports complement academic literature.
C9	Healthcare	Medium	6–12 months	Academic + News + Technical documentation	Tests combined effect of procedural and evidence-based sources in clinical context.

C10	Historical	Low	> 6 months (domain-adjusted)	Academic only	Baseline condition for low-volatility domain where freshness is expected to matter less (Metzler et al., 2021).
C11	Historical	Low	> 6 months (domain-adjusted)	Academic + News	Tests whether primary sources complement scholarly interpretation in stable domains.
C12	Historical	Low	> 6 months (domain-adjusted)	Academic + News + Technical documentation	Tests whether source diversity improves coverage without harming precision in low-volatility domains.

3.4.3 System Configuration

The RAG system is built from open-source components to ensure that the experimental results can be replicated. Proprietary systems introduce black-box retrieval and generation behaviors that cannot be controlled or verified. Open-source implementations allow every parameter to be documented and fixed. Table 3.3 below—the performance metrics table—and Table 3.5 below detail the configuration.

Dense Passage Retrieval with the all-mpnet-base-v2 sentence transformer is used for retrieval. This model was chosen because it performs consistently across multiple domain types without requiring domain-specific fine-tuning. If a domain-adapted retrieval model were used, it would be unclear whether performance differences between domains reflected domain volatility characteristics or retrieval model specialization. Retrieval depth is set at $k = 5$, following the configuration used in the system configuration table. A sensitivity analysis comparing $k = 3$ and $k = 5$ will be conducted during pilot testing to confirm that primary metric results are stable across these retrieval depths before the main experiment begins.

T5-large handles generation. It is open-source, has been benchmarked extensively on knowledge-intensive tasks, and is small enough to run efficiently across the large number of experimental conditions required while retaining sufficient capacity for complex generations. The

prompt template is fixed across all conditions. Prompt engineering affects generation quality substantially and allowing it to vary would confound the results.

Table 3.5: RAG System Configuration and Justification for Each Design Choice

Component (identified in LR)	Configuration Used	Reason for This Choice
Retrieval model	Dense Passage Retrieval (DPR) with sentence-transformers/all-mpnet-base-v2	DPR provides a well-validated dense retrieval baseline that does not require domain-specific fine-tuning. Using the same embedding model across all domains ensures that any performance differences reflect dataset characteristics, not retrieval model adaptation effects (Karpukhin et al., 2020). <i>LR basis: LR identified DPR as the standard dense retrieval architecture (Lewis et al., 2021; Karpukhin et al., 2020).</i> <i>LR basis: LR found $k=5$ balances recall against noise in retrieval across domain types (Ram et al., 2023).</i>
Generation model	T5-large (770 M parameters) via Hugging Face Transformers	T5-large is open-source, reproducible, and has been benchmarked extensively on knowledge-intensive generation tasks. Its parameter scale is sufficient for complex generation without requiring infrastructure that would make replication difficult (Raffel et al., 2020). <i>LR basis: LR benchmarked T5-large as an open-source model with extensive knowledge-intensive generation evaluations (Raffel et al., 2020).</i>
Retrieval depth (k)	Top-5 with cosine similarity threshold ≥ 0.65	Retrieving five passages balances recall—ensuring relevant documents are in the candidate set—against noise from low-relevance results. The similarity threshold prevents very weak matches from entering the generation context. Ram et al. (2023) found that similar settings produced stable results across domain types.
Chunk size	512 tokens with 50-token overlap	512-token chunks align with DPR's training context window, and a 50-token overlap prevents key information from being severed at chunk boundaries. This follows the configuration used in Lewis et al. (2021) for consistency with foundational RAG work. <i>LR basis: LR showed 512-token chunks align with DPR's training context window (Lewis et al., 2021).</i>

Reranking	Cross-encoder reranker applied uniformly across all conditions	A reranker is included to improve retrieval quality, but it is held constant across every condition. This means any performance differences between conditions cannot be attributed to the reranking step. <i>LR basis: LR used cross-encoder rerankers consistently to isolate knowledge-base effects from retrieval model effects.</i>
Prompt template	Fixed, standardized template for all queries across all conditions	Prompt engineering can substantially alter generation outputs. Using the same template throughout means the study measures how the document collection affects responses, not how prompt variation does. <i>LR basis: LR established that prompt variation substantially alters generation outputs, requiring fixed templates for controlled experiments.</i>
Vector store	FAISS with exact inner-product search	FAISS provides deterministic retrieval results. Approximate nearest-neighbour search introduces stochastic variation; exact search eliminates that source of noise from the experimental results. <i>LR basis: LR flagged approximate search as a source of stochastic variance; exact search eliminates this confound (Lewis et al., 2021).</i>
Evaluation: automated	BERTScore, ROUGE-L, METEOR, DeBERTa-based NLI entailment	These tools cover complementary dimensions of quality. BERTScore captures semantic similarity; ROUGE-L measures content recall; NLI entailment checking identifies generated claims unsupported by retrieved passages. <i>LR basis: LR established that 28% of hallucinations originate from retrieved content (Honovich et al., 2022), requiring NLI entailment checking rather than surface-level scoring alone.</i>
Evaluation: human	Two independent raters, 5-point Likert scale, Cohen's $\kappa \geq 0.70$ required	Human evaluation is restricted to a stratified 20% sample due to resource constraints. The inter-rater reliability threshold ensures that scores used in analysis reflect genuine agreement rather than rater drift. <i>LR basis: Inter-rater reliability threshold of $\kappa \geq 0.70$ reflects standard practice for human evaluation in NLP studies (Lakens, 2022); this threshold was applied in RAG evaluation contexts reviewed in the literature (Kamalloo et al., 2023).</i>

3.4.4 Performance Metrics

The performance metrics used in this study are listed and explained in Table 3.3. Three primary metrics are measured: retrieval precision, response accuracy, and hallucination rate. These were chosen because they map directly to the ways in which dataset characteristics are expected to

matter. Freshness decay should reduce retrieval precision first—as documents age, they become less relevant to current queries—and then propagate through to response accuracy and hallucination rate as the system generates responses grounded in outdated retrieved content. Source type diversity is expected to affect these metrics differently: academic sources tend to improve factual accuracy; news sources tend to improve coverage of recent events; technical documentation improves precision for domain-specific procedural queries.

Secondary metric response relevance, source attribution, and processing latency—provide additional context. They are not the primary evidence for the study's conclusions, but they help explain patterns in the primary metrics. If primary accuracy improves with diverse source types but source attribution scores do not, that suggests the accuracy improvement comes from broader coverage rather than from better-cited claims.

Two factor-level metrics—freshness score and source diversity index—operationalize the independent variables as continuous quantities rather than categorical levels. These are used in the regression and framework-construction phases to model factor-performance relationships without being constrained by the four-level categorical structure of the experimental conditions.

Table 3.3: Performance Metrics — Definition, Measurement Method, and Role in the Study

Metric	Category	What It Measures	Measurement Approach	Why It Matters for This Study
Retrieval Precision	Primary	Proportion of the top-5 retrieved documents are genuinely relevant to the query.	Precision at k=5 (P@5) and nDCG@5 computed against relevance judgements from domain experts.	Directly shows whether freshness and source type influence which documents the system surfaces. A drop in precision with older documents provides evidence for freshness effects (Thakur et al., 2021).

Response Accuracy	Primary	Whether the generated response is factually correct relative to a pre-constructed reference answer.	BERTScore against reference answers, supplemented by human expert ratings on a 5-point Likert scale.	Accuracy ties retrieval quality to what the user actually receives. Changes in accuracy across source type conditions indicate that document origin—not just retrieval mechanism—shapes response quality. BERTScore provides a semantically sensitive accuracy measure that captures meaning-level correctness beyond surface n-gram overlap (Zhang et al., 2020).
Hallucination Rate	Primary	Percentage of claims generated that are not supported by any retrieved document.	Natural Language Inference using a DeBERTa-based entailment classifier. Each claim is checked against the retrieved passage set.	Mallen et al. (2023) found that roughly 28% of hallucinations originate from poor retrieved content rather than from the generation model. Tracking hallucination rates here makes it possible to assess how dataset characteristics—particularly outdated or narrow sources—contribute to that problem. Detection uses DeBERTa-based natural language inference entailment checking (Honovich et al., 2022).
Response Relevance	Secondary	Whether the response meaningfully addresses the query, beyond factual correctness.	ROUGE-L, METEOR, and cosine similarity against reference answers.	Source type diversity may improve relevance even when accuracy is held roughly constant. This metric captures that effect. ROUGE-L measures content recall at the sequence level (Fabbri et al., 2021), while METEOR and COMET provide complementary semantic and neural evaluation measures (Rei et al., 2020).
Source Attribution	Secondary	Whether the response correctly identifies which document supported each claim.	Exact-match tracking of source identifiers between retrieved documents and generated responses.	Particularly useful for assessing whether diverse source type configurations produce responses that are transparent about their origins.

Processing Latency	Secondary	End-to-end response time in milliseconds, from query submission to response delivery.	System timer averaged across 200 queries per condition.	Diverse source collections may be larger and slower to search. Latency is tracked to assess whether performance gains come at a practical cost.
Freshness Score	Factor metric	Elapsed time in days between document publication and retrieval date, normalized to a 0–1 scale.	Computed from document metadata timestamps: (retrieval date – publication date) / max observed age.	Operationalizes freshness as a continuous variable, enabling regression analysis beyond the four categorical freshness windows.
Source Diversity Index	Factor metric	Spread of document types in the retrieved set for a given query.	Shannon entropy across source-type proportions: $H = -\sum p \log p$.	Allows source diversity to be treated as a quantity rather than a category, supporting continuous analysis of how diversity relates to performance.

3.4.5 Evaluation Protocol

Automated evaluation covers retrieval quality and response quality. BERT Score is used to compare generated responses against reference answers at the semantic level. ROUGE-L and METEOR provide complementary overlap-based measures. Hallucination detection uses a DeBERTa-based Natural Language Inference classifier that checks whether each generated claim is entailed by the retrieved passage set—a method that Mallen et al. (2023) found to be more reliable than surface-level factual verification for this kind of analysis.

Human evaluation covers a stratified 20% sample of responses across conditions. With up to 2,400 total query-response pairs, this amounts to approximately 480 responses rated by human evaluators. Two independent raters divide this work, with each rater evaluating approximately 240 responses. Responses are scored on a five-point Likert scale for accuracy and relevance. Before human scores were used in analysis, Cohen's kappa is computed for each rater pair. A threshold of $\kappa \geq 0.70$ is required; if raters fall below this threshold, a calibration session is held before scoring

continues. This is a conservative threshold, but it is appropriate given that the study's conclusions about source type effects rely partly on human quality judgements.

Topic distribution within each domain is balanced to avoid over-representation of specific subtopics that could bias performance in either direction. For the technology domain, this means distributing queries across software, hardware, artificial intelligence, and systems topics rather than concentrating on a single area. For healthcare, queries span clinical treatment, pharmacology, diagnostics, and public health. This balance reduces the risk that performance differences between source type conditions reflect topical coverage advantages rather than structural properties of the source types themselves.

3.5 Phase 3: Framework Development

Statistical analysis serves two purposes: testing whether the factors produce detectable and substantial effects on performance and building the empirical foundation that the predictive framework rests on. These are related but distinct tasks. Significance testing answers whether effects exist. Effect size and regression analysis answers how large they are and whether they can be described in a form that supports prediction.

Analysis proceeds in stages. Descriptive statistics come first—distributions, means, and standard deviations across conditions—to confirm that the data look reasonable and to check for outliers or distributional violations before inferential analysis begins. Prior to running the ANOVA, assumptions of normality and homogeneity of variance are tested using Shapiro-Wilk and Levene's tests respectively. Where violations are found, appropriate transformations or non-parametric alternatives will be applied. A two-way ANOVA then tests the main effects of freshness level and source type configuration on each primary metric, as well as their interaction. The interaction term is of particular interest. If freshness and source type effects are additive and independent, the predictive framework can treat them separately. If they interact, the framework must model their joint effects—a more complex but more accurate representation of reality.

Where the ANOVA finds significant effects, post-hoc Tukey HSD comparisons with Bonferroni correction identify which specific pairs of conditions differ. The Bonferroni correction is conservative but appropriate given the number of pairwise comparisons involved. Effect sizes are reported alongside p-values throughout, using Cohen's d for pairwise comparisons and η^2 (eta-squared) for the ANOVA factors. Effect sizes will be interpreted alongside significance values to avoid placing undue weight on statistically significant but practically negligible differences—a result with $p < 0.05$ but η^2 of 0.02 warrants a different response from the framework than one with the same significance level but η^2 of 0.28.

Multiple regression and Random Forest analysis are both applied to model factor-performance relationships. Regression produces interpretable coefficients, the kind that can be read directly into a practitioner-facing framework. Random Forest serves as a non-parametric validation layer to detect interaction patterns and non-linear effects not captured by the linear regression model; it does not replace regression but complements it. Comparing the two approaches provides a check on consistency: where regression coefficients and Random Forest feature importance scores agree on which factors matter most, confidence in those conclusions is higher. Where they diverge, the source of the discrepancy is investigated before either result is incorporated into the framework.

Ten-fold cross-validation is applied to both modelling approaches, with folds stratified to preserve proportional representation of both domain and source-type conditions in each fold. This is a stronger requirement than domain-only stratification: it ensures that the held-out fold in each iteration contains observations from all three domains and all three source-type configurations, rather than just from all domains. Cross-validation results determine how confidently the framework can be applied beyond the specific conditions tested. If prediction error is low and consistent across folds, the framework's factor-performance relationships are reliable. If error is high or variable, the framework's output must be presented with wider uncertainty bounds.

Table 3.6 below summarizes the full analysis plan, linking each analytical technique to its purpose and to how its outputs feed into the framework construction phase that follows.

Table 3.6: Statistical Analysis Plan — Technique, Purpose, and Contribution to the Framework

Stage	Technique	What It Tests or Builds	How It Feeds into the Framework
Descriptive analysis	Mean, standard deviation, distribution plots, correlation matrix across 12 conditions	Characterises performance variation before attributing causes. Identifies outlier conditions and checks distributional assumptions for ANOVA (Field, 2018).	Provides the factual summary that anchors all subsequent analysis. Without this step, patterns in the inferential analysis cannot be distinguished from data artifacts.
Two-way ANOVA	Factorial ANOVA with freshness level × source type configuration as factors; three primary metrics as dependent variables; $\alpha = 0.05$	Tests whether freshness effects, source type effects, and their interaction are statistically distinguishable from random variation within conditions (Field, 2018).	The interaction term is particularly important. If the interaction is significant, the framework must account for how freshness and source type jointly shape performance rather than treating them as independent inputs.
Post-hoc comparison	Tukey HSD with Bonferroni correction	Identifies which specific pairs of freshness levels or source type configurations differ from each other after a significant ANOVA result.	Pinpoints the boundaries between conditions where performance changes meaningfully. These boundaries become the threshold recommendations in the framework's practical guidelines.
Effect size reporting	Cohen's d for pairwise comparisons; η^2 (eta-squared) for ANOVA factors	Quantifies how large each factor's contribution is, not just whether it is statistically detectable (Lakens, 2022).	Effect sizes determine which factors deserve greater weight in the framework. A factor that is statistically significant but accounts for only 2% of variance is a minor input; one accounting for 30% is a primary driver.
Multiple regression	Ordinary least squares regression with freshness score, source diversity index, domain volatility, and interaction terms as predictors	Models the linear relationship between continuous factor values and performance metrics. Produces interpretable coefficients (James et al., 2023).	The regression equations form the first, interpretable layer of the framework. Practitioners can use them to estimate expected performance for a given dataset configuration without running experiments.

Non-linear curve fitting	Polynomial regression; exponential decay modelling for freshness effects	Tests whether freshness decay follows a predictable curve that can be parameterized by domain volatility.	If decay follows a consistent shape, the framework can include decay curves that let practitioners project performance forward in time from a baseline measurement.
Random Forest analysis	Ensemble of 500 decision trees with grid-searched hyperparameters	Captures non-linear relationships and factor interactions without requiring pre-specified functional forms. Feature importance scores identify which factors matter most.	Complements the regression layer. Where regression coefficients and feature importance scores converge, confidence in the framework's factor weightings is higher. Where they diverge, additional investigation is warranted.
10-fold cross-validation	Stratified by domain; repeated for each modelling approach	Assesses whether factor-performance relationships established from the experimental data generalize to unseen conditions.	Cross-validation results determine how confidently the framework can be applied to new domains or configurations beyond those explicitly tested. Prediction errors are bound inform how the framework's uncertainty should be communicated to practitioners.

3.6 Phase 4: Framework Validation and Benchmarking

Phase 4 is where the statistical results become a framework. The distinction matters. Statistical results describe what was found in the experimental conditions. A framework translates those findings into a form that someone who did not run the experiments can use when making decisions about their own RAG system. As defined in Section 3.1, the framework predicts expected performance ranges, not exact response content or individual model behavior.

Figure 3.2 below illustrates the conceptual architecture of the predictive framework. The flow moves from raw dataset characteristics through a quantification step, into an interaction modelling layer, and finally into performance estimation outputs that support deployment decisions.

Figure 3.2: Predictive Framework Architecture — From Dataset Factors to Decision Support





3.6.1 Framework Components

The framework consists of six components that together translate experimental findings into a usable decision-support tool. Each component is listed here to make the framework's structure

explicit, since these elements are developed progressively within Phase 4 and could otherwise remain implicit in the narrative.

The first component is factor definitions: precise, operationalized descriptions of the two primary factors—freshness and source type, including how each is measured and how domain adjustment is handled. These definitions are the foundation on which everything else rests; a practitioner applying the framework to their own dataset must be able to map their data to these definitions.

The second component is quantification methods: the procedures for computing freshness score and source diversity index as continuous variables from document metadata. The source diversity index uses Shannon entropy across source-type proportions. It is worth noting a limitation of this measure: Shannon entropy captures the spread of source types in a retrieved set but does not measure quality variation within any given type. A collection of uniformly low-quality news articles will receive the same diversity score as a collection of mixed high- and low-quality news articles. This limitation does not undermine the index's utility for comparing across conditions, but it should be kept in mind when applying the framework to collections where within-type quality varies substantially.

The third component is the interaction modelling layer: the regression equations and Random Forest feature importance analysis that describe how freshness and source type jointly affect each primary metric. This layer is where the framework captures the nuance that simple heuristics miss—for instance, that source type diversity matters more in high-volatility domains than in low-volatility ones.

The fourth component is performance estimation equations: the specific formulas, derived from regression analysis and cross-validated, that translate input factor values into expected performance ranges with confidence intervals. These are expressed in a form that can be used without running the full experiment—an organization inputs their dataset's average freshness score and source diversity index, and the equation returns an estimated performance range for their domain type.

The fifth component is benchmark tables: pre-computed performance ranges for the twelve experimental conditions, presented with 95% confidence intervals. These serve as ready references for common configurations and allow organizations to locate their situation within the framework without needing to use the estimation equations directly.

The sixth component is boundary conditions: documented circumstances where the framework's estimates are less reliable. These include domains with volatility profiles that differ substantially from the three tested, collections where source classification is ambiguous, and query distributions heavily weighted toward one complexity type. Being explicit about boundary conditions is what distinguishes a framework that is useful from one that is merely confident.

3.6.2 Validation

Validation tests whether the framework's estimates hold against conditions that were not used to build it. A portion of the experimental data—conditions held out during the regression and cross-validation stages—is reserved for this purpose. If the framework's predicted performance ranges cover the observed results in the held-out conditions at the specified confidence level, the validation is considered satisfactory. Where predictions are missed, the discrepancy is investigated to determine whether it reflects a genuine boundary condition or a structural gap in the framework.

The framework demonstrates feasibility of factor-based performance prediction using two representative dataset-level factors. It does not claim to be exhaustive across all possible dataset properties, and extending it to additional factors—document length, citation density, authoritativeness scores—is an explicit direction for future work. The contribution is a proof-of-concept that systematic, evidence-based prediction from dataset characteristics is achievable, together with the empirical relationships and validated benchmarks that make that prediction possible.

3.7 Validity and Ethical Considerations

Internal validity depends primarily on the design decision to hold architecture constant across all conditions. Because the retrieval model, generation model, and prompt template do not change, performance differences between conditions cannot be attributed to architectural factors. Two remaining threats to internal validity are worth acknowledging. First, documents in different freshness windows might differ in ways other than age, for instance, in topic density, writing style, or authorial quality. This is addressed by the quality assurance procedures in Phase 1 and by the domain expert verification step. Second, topic distribution could skew performance if certain subtopics are over-represented in particular freshness windows. This is addressed by the topic balancing procedure described in Section 3.4.5, which distributes queries evenly across subtopics within each domain. Neither threat can be fully eliminated, but both are systematically managed. External validity is partial. The framework is validated across three domains and three source type configurations, which represents a reasonable breadth for an academic study. However, it does not cover all domains where RAG systems are deployed, and the specific documents collected here reflect what was available from the listed sources between 2015 and 2025. Organizations working in substantially different domains, or with document collections assembled from very different source types, should treat the framework's estimates as indicative rather than prescriptive until validated against their own data.

The study uses only publicly available documents. No personally identifiable information is collected or processed. Human evaluators participate voluntarily, are informed of the rating criteria and the study's purpose before beginning and provide consent to have their ratings used in the analysis. No other human participants are involved.

3.8 Chapter Summary

This chapter described the reasoning behind a four-phase methodology designed to investigate whether dataset freshness and source type can support a predictive framework for RAG system performance. Each phase corresponds directly to one of the four research objectives set out in Table 3.1. A predictive framework, as defined in Section 3.1, is a structured, factor-based system

that quantifies dataset characteristics, models their interactions, and translates those relationships into performance estimation ranges and deployment guidelines. The framework predicts expected ranges for retrieval precision, response accuracy, and hallucination rate—not individual query outcomes or exact model behavior.

The design is quasi-experimental and domain adjusted. Because certain freshness windows are not meaningful in low-volatility domains, a full $4 \times 3 \times 3$ factorial cross is replaced by twelve domain-appropriate conditions that preserve conceptual equivalence through the use of both categorical windows and a continuous freshness score. Within each domain, the same query bank is used across all source-type conditions, which isolates source effects cleanly. Time-sensitive queries are classified by verified ground-truth change across freshness windows, not by assumption.

Statistical analysis proceeds through ANOVA—with pre-tested assumptions—to regression and Random Forest modelling, with cross-validation stratified by both domain and source-type condition. Effect sizes are reported alongside significance values throughout, and the Random Forest serves as a non-parametric validation layer rather than a replacement for regression. The framework's six components—factor definitions, quantification methods, interaction modelling layer, performance estimation equations, benchmark tables, and boundary conditions—are constructed from these analytical outputs in Phase 4 and validated against held-out conditions.

Two limitations on metric scope are explicitly stated. The source diversity index captures spread across source types but not quality variation within a type. And the framework demonstrates feasibility using two representative dataset-level factors; it does not claim to account for all properties that could influence RAG performance. These are starting points, not endpoints.

Chapter 4 will present the results of the controlled experiments, beginning with descriptive statistics across the twelve conditions and proceeding through factorial analysis, regression modelling, and the construction and validation of the framework described here.

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APPENDIX: KEY FINDINGS SUMMARY OF CITED LITERATURE

Table A.1 below summarizes the research focus, methodology, key findings, and limitations.

Table A.1: Key Findings Summary of All Cited Literature

Citation	Research Focus	Methodology	Key Findings	Limitations & Relevance to This Study
Asai et al. (2023)	Self-corrective RAG with reflection tokens	Introduced SELF-RAG; fine-tuned LLaMA-2 to output special tokens that judge retrieved passages before using them	Self-reflective components reduced hallucinations by approximately 32% over standard RAG. Model can selectively ignore unhelpful passages, improving factual grounding.	Adds computational overhead. Does not address knowledge-base composition or update scheduling. Improvements depend on quality of the retrieval pool.
Chen et al. (2022)	Explainability and source diversity in RAG outputs	Experimental evaluation tracking which retrieved documents influenced generation; user trust ratings across varied source configurations	Source type diversity increased explainability ratings by 41%. Users more readily verified claims when responses cited multiple source types. Official documentation dominated technical queries.	Limited exploration of how freshness interacts with explainability. User sample may not generalise across demographics or professional contexts.
Creswell & Creswell (2018)	Research design methodology for social and applied sciences	Methodological textbook; synthesises qualitative, quantitative, and mixed-methods approaches with philosophical underpinnings	Positivist, deductive approaches are appropriate when testing hypotheses about measurable relationships. Quasi-experimental designs suit research where full randomisation is not feasible.	Not an empirical study. Provides frameworks rather than domain-specific guidance for RAG or NLP research.
Gao et al. (2023)	Zero-shot dense retrieval without relevance labels	Multi-metric evaluation of retrieval precision across heterogeneous source collections using HyDE (Hypothetical Document Embeddings)	Academic sources improved factual precision; news sources enhanced contemporary relevance; technical documentation improved domain-specific accuracy. Retrieval precision varied substantially by source type.	Does not systematically investigate the interaction between source type and temporal freshness. Evaluation is focused on retrieval, not end-to-end generation quality.
Hofstätter et al. (2023)	Efficient and effective retrieval-augmented text generation	Comparative study of domain-specific statistical prediction models for RAG performance; FiD-Light architecture evaluation across multiple domains	Domain-specific statistical models explained up to 50% of performance variance within domains. Hybrid ensemble approaches achieved 81% domain-specific accuracy. Models do not transfer across domains without retraining.	Requires separate models per domain, increasing development overhead. Limited factor isolation: does not decompose contributions of freshness versus source type independently.
Honovich et al. (2022)	Factual consistency	Developed TRUE benchmark; evaluated multiple	Approximately 28% of hallucinations in RAG-style systems originate from the	Evaluation concentrated on summarisation tasks; generalisability to open-domain QA

Citation	Research Focus	Methodology	Key Findings	Limitations & Relevance to This Study
	evaluation for NLG systems	NLI-based factuality scorers against human annotations across summarisation and dialogue tasks	retrieved content itself, not the generation model. NLI entailment checking outperforms surface-level overlap metrics for detecting unsupported claims.	and long-form RAG generation requires additional validation.
Izacard et al. (2023)	Few-shot learning with retrieval-augmented models (Atlas)	Benchmarked Atlas across 20+ knowledge-intensive tasks; varied retrieval recency and measured performance on time-sensitive versus time-neutral queries	Recent sources improved few-shot performance by approximately 17% on time-sensitive tasks. Freshness affects not only factual recall but also contextual reasoning ability. Atlas outperformed GPT-3 on several tasks with far fewer parameters.	Does not develop a predictive model for determining optimal freshness levels for a given application. Few-shot setting may not reflect production RAG deployments with larger retrieval corpora.
Kamalloo et al. (2023)	Evaluating QA performance in the era of LLMs	Trained random forest classifiers using dataset-level features (document age, source type distributions, collection size, domain) to predict whether queries would be answered correctly	Achieved 71% accuracy in predicting correct query answering. Document age and source composition ranked among top predictive features. Dataset-level factors are strong performance predictors.	Requires large labeled training datasets. Produces binary predictions (correct/incorrect) rather than continuous performance estimates. Does not generalise well to new configurations without retraining.
Karpukhin et al. (2020)	Dense Passage Retrieval (DPR) for open-domain QA	Trained dual-encoder model using in-batch negatives on Natural Questions and TriviaQA; compared against BM25 and hybrid approaches	DPR substantially outperformed BM25 on open-domain QA benchmarks. Dense vector representations enable semantic matching beyond keyword overlap. Became standard retrieval component for subsequent RAG research.	Domain-adapted models improve performance in specialised contexts but require additional training data. Embeddings trained on one domain may not transfer optimally to others.
Kasai et al. (2023)	Real-time QA: measuring RAG performance on current-events questions	Longitudinal benchmark tracking RAG accuracy on weekly questions; classified information decay into gradual, threshold, and cyclical patterns across domains	Technology exhibits gradual decay; financial information shows threshold patterns; medical research displays cyclical patterns. Domain volatility characteristics are measurably distinct and require domain-adjusted update strategies.	Primarily focused on news-driven queries. Does not develop a predictive framework for new domains or provide generalisation guidance beyond the domains examined.
Lewis et al. (2021)	Foundational RAG architecture: retrieval-augmented generation for knowledge-intensive NLP	Combined DPR retrieval with BART generation; evaluated on knowledge-intensive benchmarks using Wikipedia as sole knowledge source	RAG improved factual accuracy by 9–19% depending on task. External retrieval reduces hallucinations versus purely parametric generation. Initial Wikipedia indexing required approximately 42 GPU hours.	Limited to Wikipedia as knowledge source; does not examine diverse source types. Does not address temporal freshness or knowledge-base maintenance strategies.

Citation	Research Focus	Methodology	Key Findings	Limitations & Relevance to This Study
Mallen et al. (2023)	When to trust language models: parametric versus non-parametric memory	Developed evaluation framework isolating retrieval quality effects from generation quality; analysed 78% of implementation failures across diverse RAG deployments	78% of implementation failures stem from dataset problems rather than algorithmic limitations. 28% of hallucinations originate from retrieved content. Entity-mention timestamps improve temporal accuracy by 23% over document-level dates.	Temporal performance aspects not fully modelled. Does not provide predictive guidance for update scheduling or source selection. Entity timestamp tracking requires substantial preprocessing overhead.
Metzler et al. (2021)	Temporal decay modelling for information retrieval systems	Developed mathematical decay functions calibrated per source type; applied to document weighting during retrieval; compared decay parameter sets across news and academic domains	News articles assigned 30-day half-life decay; academic papers assigned 2-year half-life. Applying decay weights improved temporal accuracy by 18% on time-sensitive queries.	Decay parameters require domain-specific calibration; do not transfer between domains. Approach focuses on retrieval weighting, not on predicting how overall RAG performance changes as knowledge bases age.
Petroni et al. (2021)	KILT: benchmark for knowledge-intensive language tasks	Created standardised benchmark across five task types; developed regression models explaining performance variance based on knowledge-base properties including source diversity, collection size, and domain	Regression models explained 32% of performance variance. Knowledge-base composition significantly affects task performance. Established baseline for knowledge-intensive evaluation.	Retrospective analysis requires historical performance data; cannot forecast performance for new configurations. Explained variance is limited, indicating important unmeasured factors.
Raffel et al. (2020)	T5: unified text-to-text transfer transformer	Pre-trained and fine-tuned T5 across a comprehensive suite of NLP tasks using a unified text-to-text format; ablation studies across model scales from 60M to 11B parameters	T5-large achieves competitive performance on knowledge-intensive generation tasks. Unified text-to-text format enables consistent fine-tuning across diverse tasks. Scales efficiently with parameter count.	Static parametric knowledge without retrieval integration. Performance on rapidly changing knowledge degrades without retraining. Larger scales require significant computational resources.
Ram et al. (2023)	In-context retrieval-augmented language models	Evaluated in-context RAG across multiple task types; measured retrieval-to-generation latency ratios; conducted ablation	Retrieval accounts for 40–60% of total inference time. $k=5$ retrieval balances recall and noise across domain types. $nDCG@10$ correlates strongly with generation quality, supporting retrieval ranking importance.	In-context approach constrained by context window size. Does not address temporal freshness effects on long-horizon performance. Results may not generalise to very large corpora.

Citation	Research Focus	Methodology	Key Findings	Limitations & Relevance to This Study
		on retrieval depth k and chunk size		
Shadish et al. (2002)	Experimental and quasi-experimental design for causal inference	Methodological textbook; reviews design types, threats to validity, and statistical approaches for research contexts where full randomisation is not feasible	Quasi-experimental designs are appropriate and defensible when treatment assignment cannot be randomised, provided that confounds are systematically controlled. Careful matching and control procedures can compensate for non-random assignment.	Not an empirical study. Provides general design principles; domain-specific application (e.g., to temporal information retrieval) requires researcher judgement.
Xiong et al. (2024)	Source diversity, credibility, and user trust in medical RAG systems	Benchmarked RAG pipelines across medical question-answering tasks; evaluated how source type and document recency affected answer accuracy, user trust, and maintenance overhead in clinical settings	Source diversity improved user trust ratings in RAG outputs. Academic and clinical sources outperformed general web content for factual accuracy. Production deployments required sustained maintenance effort to preserve retrieval quality over time. Source type composition significantly affected perceived credibility across user groups.	Focused on medical domain; generalisability to technology or historical research requires validation. Does not examine temporal decay in detail. Source diversity effects may differ in non-clinical contexts.
Thakur et al. (2021)	BEIR: heterogeneous benchmark for zero-shot retrieval evaluation	Compiled 18 diverse retrieval datasets spanning multiple domains and task types; evaluated multiple retrieval models in zero-shot settings	Retrieval precision varied substantially across domains (P@10: 0.42 to 0.78). Models optimised for one domain often performed poorly on others. Established that domain-specific evaluation is essential for realistic performance assessment.	Evaluates only retrieval, not end-to-end RAG generation quality. Zero-shot setting does not reflect all production deployment scenarios. Relevance judgements may conflate topical relevance with temporal currency.
Wang et al. (2022)	Impact of source quality and type on RAG performance	Comparative analysis of parametric versus non-parametric knowledge across varied document collections; quality scoring using composite signals (citation counts, venue prestige, structural features)	High-quality academic sources improved factual accuracy by 27%, but improvements plateaued beyond a threshold. Expert curation contributed 30–40% of implementation effort. Non-linear relationship between source quality and performance.	Does not examine temporal freshness effects. Non-linear relationships are documented but not modelled predictively. Quality metrics correlate with publication delay, creating a quality–timeliness tradeoff not fully resolved.
Yu et al. (2023)	Generate rather than retrieve: Large language models as context learners	Comparative evaluation of LLM-generated context versus retrieved passages across multiple QA tasks; measured accuracy and relevance	LLMs can generate relevant context that partially substitutes for retrieved documents in closed-book settings. Performance comparable to retrieval-augmented approaches on certain QA benchmarks, though with higher hallucination	Does not address knowledge base update strategies or temporal freshness. Focuses on replacing rather than improving retrieval. Results may not generalise to domains where retrieved content is essential for factual accuracy.

Citation	Research Focus	Methodology	Key Findings	Limitations & Relevance to This Study
		tradeoffs when substituting generation for retrieval	risk. Demonstrates that knowledge source type (parametric vs. retrieved) affects output quality.	
Zhang et al. (2022)	Temporal performance decay in RAG systems across domains	Longitudinal study tracking RAG accuracy monthly over six months as knowledge bases aged without updates; domain comparison across technology, healthcare, and historical research	Performance in technology domains declined 2.3% per month without updates. Decay was negligible in historical research over the same period. Different domains exhibit fundamentally different decay rates requiring domain-adjusted maintenance strategies.	Does not build predictive models for forecasting when performance will fall below acceptable thresholds. Limited to three domains. Decay functions are documented but not parameterised for transfer to new contexts.